

Attention predicts soft-law focus in Antarctic Treaty governance

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Governing a global commons requires actors with different concerns to assemble a shared agenda. We reconstruct the Antarctic Treaty System's documentary agenda since 1961 as a space of concerns, where concerns lie close when the same actors specialize in both. All concerns belong to one map, but they concentrate locally more than expected from broad actor portfolios and widely held concerns alone. Actors occupy this space unevenly and add new specializations near earlier portfolios. In retrospective tests, papers about the target and nearby categories improve predictions of the focus of non-binding Resolutions but provide no clear gain for Decisions or potentially binding Measures. Measures instead concentrate in recurring area-management work and most cite earlier formal instruments. The concern space links distributed attention to agenda structure and separates institutional change from continuity in potentially binding output.

Keywords: agenda formation, consensus governance, global commons, environmental governance, space of concerns, relational analysis, Antarctic Treaty

Highlights

- Shuffled comparisons reproduce overall connectivity but not local concentrations
- Actor positions differ, and new specializations occur more often near earlier portfolios
- Prior-meeting attention improves retrospective Resolution category predictions
- Measures cluster in recurring work and most cite earlier formal instruments

1 Introduction

In August 2026, average sea-surface temperature across the extra-polar oceans reached its highest daily value in the observational record [29]. The measurement stops at 60°S, but the processes moving heat through the ocean do not. As United Nations Secretary-General António Guterres put it after visiting the continent, “What happens in Antarctica doesn't stay in Antarctica” [17]. Southern Ocean circulation strongly influences global heat storage, anthropogenic carbon uptake, overturning circulation, and sea level [14, 21]. Antarctica therefore sits near the center of a climate system whose consequences extend far beyond the continent. Yet it has no permanent population and is governed not by a single sovereign authority, but through agreements among states [7, 23]. This makes Antarctica a continent-scale problem of collective governance: participants must respond to shared environmental change while accommodating distinct national priorities.

Because adopted texts record where participants reached agreement rather than the process that made agreement possible, collective governance can appear more settled in retrospect than it was. Formal outcomes reveal less about how participants divided limited attention among competing problems, balanced common goals against national constraints, or carried earlier commitments into a negotiation. They compress the shifts in attention and contestation that precede agreement [3, 33, 12]. Before an institution can decide, participants must first determine which problems

receive documentary effort. We examine this process before adoption and ask how attention is organized across actors and concerns, how actors extend their portfolios through time, and when that organization appears in formal output.

Attention moves through an institution as participants return to established problems and take up new ones from positions built through their earlier work. Agenda studies show how attention rises and falls through time [4, 22, 15], while institutional-performance research distinguishes these processes from policy outputs and outcomes [18, 10]. Governance studies connect actors, organizations, and agreements [8, 26, 25]. Shared actors provide another observable relation: two concerns are connected when the same participants devote a disproportionate share of their documentary effort to both. If later work develops from earlier portfolios, new specializations should appear near existing ones rather than uniformly across the agenda. Relatedness measures describe a comparable local process when firms, regions, and researchers move among products, industries, technologies, and research fields [19, 20, 28, 6, 16]. Treating the agenda itself as a relational object allows its collective organization and the paths of individual actors to be studied in the same representation.

Consensus makes the organization of attention before adoption especially difficult to observe. Without roll-call votes, adoption indicates that no participant blocked a text but not how strongly each participant supported it, and agreement can be reached by narrowing or weakening what the text requires [32, 12]. Adopted records also contain outputs with different legal force, making it difficult to infer one process from formal texts treated as equivalent outcomes [1]. Studying attention before agreement therefore requires repeated identifiable actors, documents dated before adoption, concern categories that can be compared through time, and formal outputs whose legal status can be distinguished.

The Antarctic Treaty System (ATS) makes this pre-adoption process unusually observable because its consensus rules suppress roll-call signals while its archive preserves the documentary sequence preceding agreement. The 1959 Treaty established peace and scientific cooperation, while later agendas came to include climate change, expanding tourism, conservation demands, and geopolitical pressures [9, 27, 11, 7]. Consultative Meetings have governed activity across Antarctica by consensus since 1961

[23, 35]. Participants submit papers that raise problems, present evidence, and propose action before meetings adopt formal texts [24, 31], and the archive provides a common retrospective concern taxonomy applied across more than six decades. Since 1995, the adopted record further distinguishes non-binding Resolutions, organizational Decisions, and Measures that can become legally binding after the required approvals. The value of this case lies in that conjunction: a continent-scale governance system, repeated actors, pre-adoption documents, comparable concerns, and outputs with different legal consequences.

We analyse 6,573 working papers and information documents submitted by 66 state and institutional actors between 1961 and 2025. Preparing a paper requires time and staff, making the distribution of this effort informative even though the documents cannot distinguish whether it reflects strategy, mandate, expertise, or routine reporting. We construct a *space of concerns* in which two concerns lie close when the same actors devote an above-average share of their papers to both. This distance represents shared specialization rather than semantic similarity, collaboration, or a causal pathway. We use it to test whether relations among concerns contain local organization beyond differences in actors' breadth of activity and concerns' prevalence, and whether actors add new specializations near their earlier portfolios. We then make sequential retrospective predictions at successive meetings, fitting each model on earlier outputs and asking whether papers about a target category and its nearby concerns improve predictions of the official categories addressed by Measures, Decisions, and Resolutions.

Our results reveal that a common agenda can be connected without making its actors interchangeable. All 45 concerns belong to one connected space, and broad actor portfolios together with widely held concerns reproduce most of that connectivity, but not its local concentrations. Actors occupy the space unevenly and add new specializations disproportionately near their earlier portfolios. The distribution of documentary attention also contains information about formal output, although the association differs across instruments. Papers about a target category and its neighbours improve retrospective predictions of the categories addressed by non-binding Resolutions in 15 of 19 meetings, and the improvement persists when the predictors use only papers from the preceding meeting. The same information provides no clear gain for organizational Decisions or potentially binding Measures. Measures concentrate in recurring Antarctic Specially Protected Area management and mostly cite earlier formal instruments. This joint pattern connects processes usually studied apart: individual specialization develops locally, and the distribution of collective documentary attention improves retrospective predictions of Resolution category focus, whereas potentially binding output retains stronger continuity with the formal record.

The documents impose a clear boundary on what this structure means because documentary effort does not reveal preferences, bargaining, influence, implementation, or environmental change. Nor do the prediction results establish that attention causes a category to appear in an adopted text. Within those bounds, adopted texts show where agreement settled; the space of concerns shows how documentary attention was organized before agreement.

2 Results

The Antarctic Treaty System governs activity across Antarctica through an agenda that extends far beyond the Treaty's founding articles [23, 7]. We reconstruct that agenda from 6,573 papers submitted by 66 state and institutional actors between 1961 and 2025. Each of the 45 circles represents a concern in the Secretariat's paper taxonomy. Two concerns lie close when the same actors specialize in both. We measure this relation with the lower of the two conditional probabilities, ϕ (eqs. (3) and (4)). The map records shared specialization. It does not measure semantic similarity, collaboration, or causal influence.

All 45 concerns belong to one connected map with distinct local concentrations. Only 39 of 990 concern pairs have a shared-specialization score of zero, so every concern connects directly or indirectly to every other concern (fig. 1b). Shuffled maps that preserve each actor's number of specializations and each concern's number of specialized actors also connect every concern and reproduce most of the overall prevalence of positive relations. They do not reproduce how strongly relations concentrate within the displayed regions. Weighted modularity, which summarizes that concentration, is $Q = 0.090$ in the observed map and 0.052 at the null median (95% interval [0.044, 0.063], $p = 0.001$; section A.9).

The displayed regions in fig. 1a show local differentiation within this connected map. Procedural and scientific work forms a broad core. Environmental management, tourism and safety, and resources and planning form denser concentrations around it. Specialized actor sets overlap across their boundaries. Inspections, area protection, and environmental protection bridge the displayed regions. Measures of overall reach and cross-region bridging distinguish widely connected concerns from bridges between regions (fig. 1c,d), while paper volume follows a different hierarchy. Counts identify where attention accumulates. Shared specialization identifies how concerns share actor portfolios. Checks on labels, the range of each actor's specializations, and actor removal leave this distinction intact (sections A.3, A.7 and A.10).

2.1 Actors develop portfolios locally

Although the map connects all concerns, actors do not hold the same portfolios. Australia spans procedural, environmental, resource, cooperation, and reporting work, whereas the International Association of Antarctica Tour Operators concentrates on access, tourism, and safety. Russia centers on the procedural core with extensions into resources and environmental concerns (fig. 2a–d). Actors joining in 1991–2010 also build their first-15-year portfolios around less connected concerns: the actors specializing in those concerns have narrower full-record portfolios, and the concerns have fewer neighbours than the opening-window benchmark (sections A.6 and B.6 and table S2). This retrospective comparison does not identify why those opening-window positions differ.

Across actors, portfolios develop locally. New specializations occur beside concerns an actor already holds, as Australia illustrates across successive five-meeting windows (fig. 2e–g). Across the archive, the rate falls from 9.4% in the nearest distance decile to 2.7% in the farthest. After

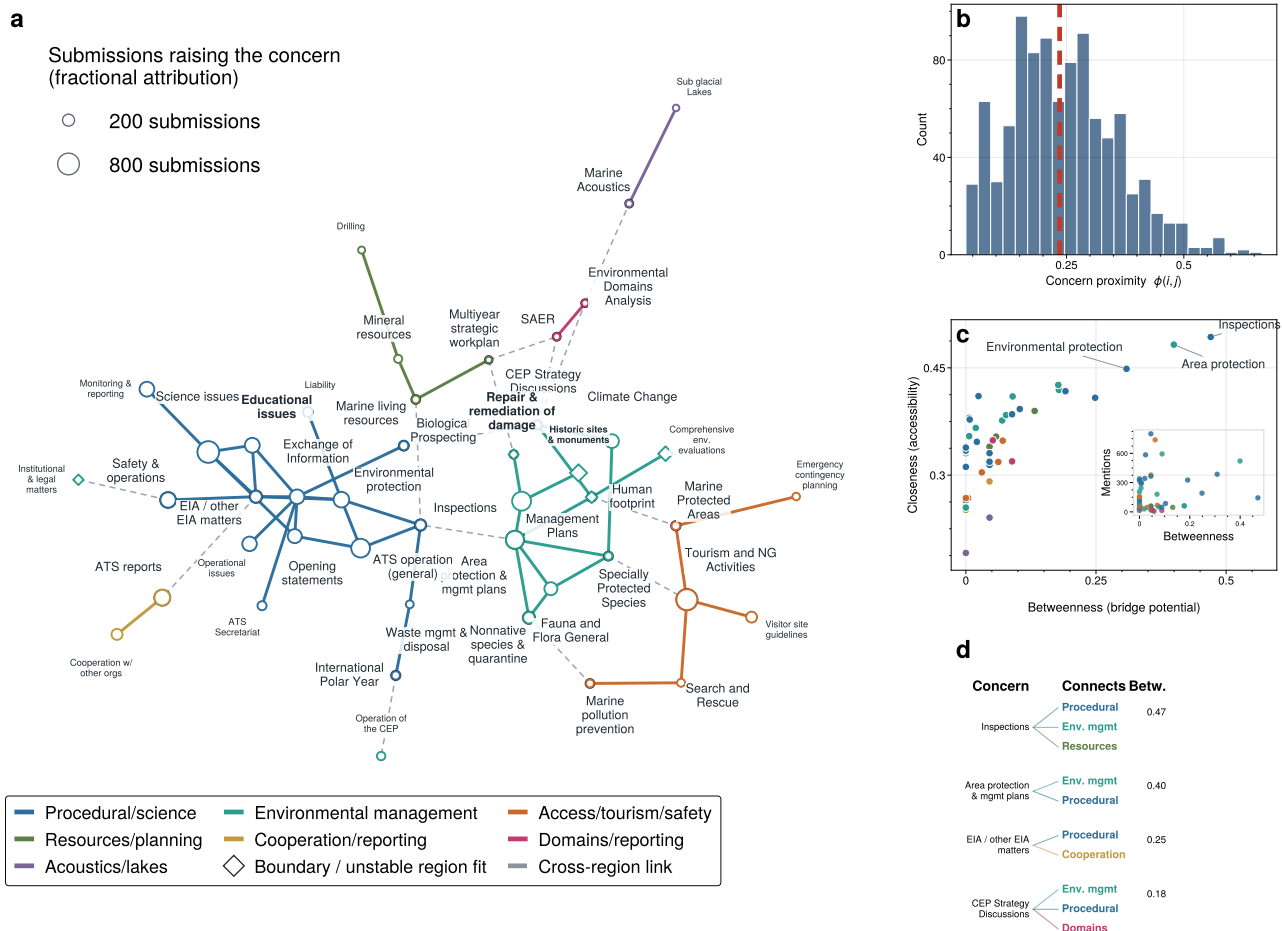


Figure 1: A common agenda contains distinct concentrations. (a) Concern map from 6,573 papers by 66 actors across 45 Secretariat concerns, 1961–2025. Lines join concerns in which the same actors specialize, and circle area gives paper count. Colours show one seven-region reading, and dashed lines cross its boundaries. Diamonds mark concerns assigned as often to another region as to the displayed one across bootstrap resamples. (b) Distribution of all 990 shared-specialization scores. (c) Overall reach across the map (closeness) and frequency on shortest paths (betweenness), with paper counts inset. (d) Concerns most often on shortest paths and their lines across regions. Sections A.2 and A.9 describe the construction and checks.

accounting for prior popularity, each 0.1 increase in distance corresponds to 18.3% lower odds of specialization (OR 0.817, 95% interval [0.783,0.853]; fig. 2h). Alternative windows, map constructions, and entry definitions preserve this direction (sections A.7 and B.7 and table S6). The result describes a pattern of portfolio development, not its cause.

2.2 Documentary attention improves retrospective Resolution prediction

Documentary attention helps locate the concerns addressed by non-binding Resolutions. We align the paper taxonomy with the 15 official instrument categories and compare the categories addressed by earlier outputs with papers about the target and nearby categories (sections A.11 and A.12). Across 19 retrospective predictions evaluated in sequence, papers about the target and nearby categories improve the Resolution prediction in 15 meetings. Relative to the categories addressed by earlier Resolutions, the combined model lowers the mean prediction score by 0.065 (95% meeting-bootstrap interval [−0.114, −0.014]; fig. 3a). It provides no clear gain for Measures or Decisions (section B.1 and table S4).

Paper volume does not explain the improvement; the

observed structure of the concern space does. That is, it outperforms every label-shuffled map in the Resolution analysis, and the combined result remains negative in a preceding-meeting sensitivity analysis (fig. 3c; section B.2). Resolutions also span a broader set of categories than Decisions or Measures, leaving more variation for current attention to locate. These tests support a retrospective association between attention and soft-law focus. They do not show that papers cause adoption or that the full predictor set was available in real time.

The paper archive and official instrument register use different Secretariat classifications. We align the 45 paper concerns with the 15 official instrument categories while leaving every official instrument-to-category assignment unchanged. Sensitivity analyses retain the overall attention association, but reduce confidence in a separate contribution from papers about nearby categories (section B.2). We therefore treat the combined Resolution result as the central finding.

Restricting all three analyses to papers attributed to Consultative Parties at each meeting leaves all 45 concerns connected and preserves local portfolio development (OR 0.900, 95% interval [0.860,0.942]). Under the same restriction, the target-and-nearby attention model still improves Resolu-

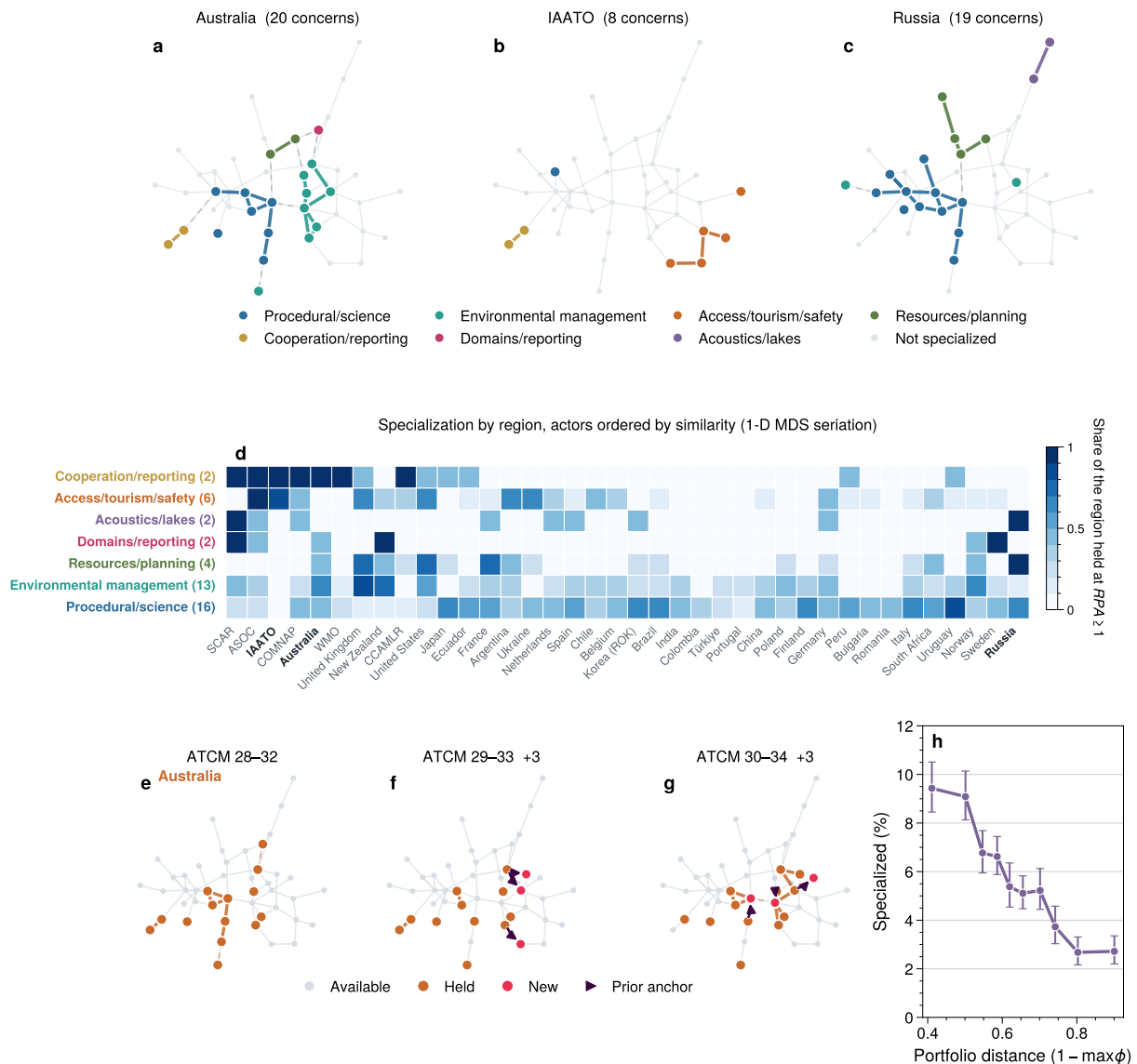


Figure 2: Actors occupy the common space differently and extend portfolios locally. (a–c) Specialization portfolios for Australia, the International Association of Antarctica Tour Operators, and Russia. (d) Regional coverage for actors with at least 50 papers. (e–g) Australia's successive five-meeting portfolios. Pink circles mark new specializations, and arrows point to their nearest earlier specializations. (h) The probability of a new specialization falls with distance from the earlier portfolio. Bars show Wilson 95% intervals. Definitions, sample sizes, and adjusted estimates appear in section A.4.

tion prediction (mean score change -0.050 , 95% meeting-bootstrap interval $[-0.100, -0.003]$), while the Decision and Measure intervals include zero (section B.5 and fig. S13).

2.3 Measures concentrate in Area protection and management

Potentially binding Measures instead concentrate in recurring area-management work. Of 277 Measures, 266 carry the official Area protection and management category, much of it involving Antarctic Specially Protected Area (ASPA) designations and management-plan revisions. ASPAs are areas designated under Annex V of the Environmental Protocol to protect outstanding environmental, scientific, historic, aesthetic, or wilderness values, or ongoing or planned scientific research [2]. Measure titles explicitly identify an ASPA in 215 cases: 160 spell out “Antarctic Specially Protected Area”, and 55 use the

acronym. A management plan appears in 230 titles and a revised management plan in 211 (fig. S11). This concentration leaves limited topical variation for prediction and may favour an output-history model based on earlier Measures. That model outperforms models based on earlier Decisions, Resolutions, or paper attention alone, and adding current attention provides no clear improvement (section B.1).

Measures also retain a formal record of that continuity. In total, 245 cite at least one earlier instrument, most often another Measure (fig. 3b). Exact citations give a conservative account because final Measures can omit intermediate steps. They identify formal reference, not a general causal path from papers to law or entry into force (section B.3 and table S5).

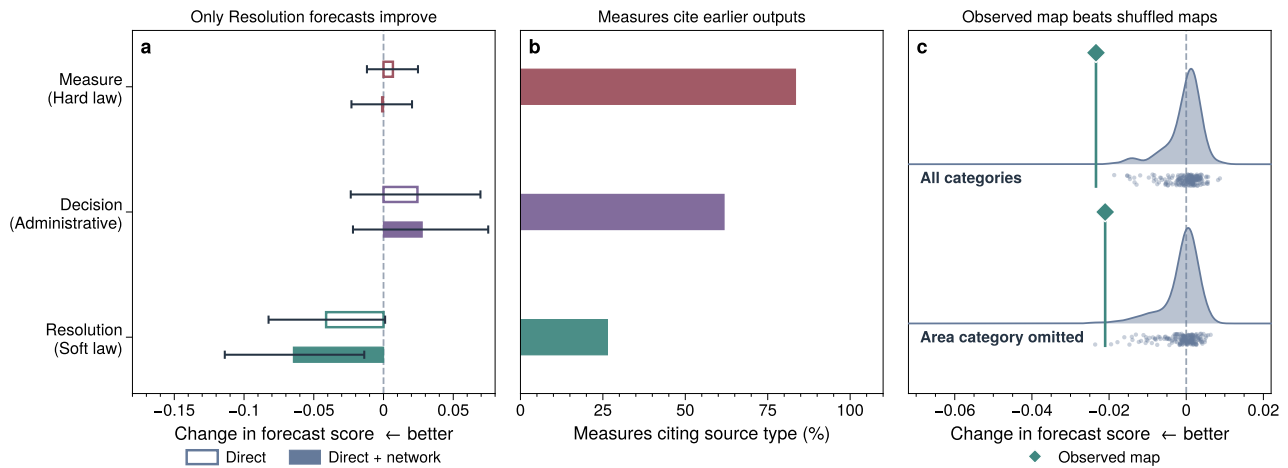


Figure 3: Attention helps locate soft-law focus, while Measures retain a formal record. (a) Change in prediction score after adding papers about the target category or the target and nearby categories to the categories addressed by earlier outputs. Lower scores indicate better prediction. Whiskers are meeting-bootstrap 95% intervals across 19 meetings. (b) Share of 277 Measures citing earlier Measures, Decisions, or Resolutions. Source types overlap. (c) Improvement from including nearby categories for the observed paper map and 200 label-shuffled maps. The lower row omits Area protection and management from scoring. Definitions and sensitivity analyses appear in sections A.12 and B.1.

3 Discussion

Governance starts taking shape before participants agree on a formal text. Papers record which problems receive documentary effort and which lines of work continue. The space of concerns brings these traces together, connecting the positions of individual actors to the agenda they build collectively.

Although all 45 concerns connect within one space, actors occupy different parts of it and usually develop new specializations near their existing portfolios. The relationship between documentary attention and formal output also differs among Resolutions, Decisions, and Measures. Documentary attention helps predict which categories non-binding Resolutions address, while earlier adopted texts remain more informative for Decisions and Measures. Recurring work on Area protection and management dominates Measures, especially Antarctic Specially Protected Area designations and management-plan revisions, which limits the topical variation available for prediction (figs. 3 and S11).

Bounded-attention research shows how institutions select problems and shift attention among them [4, 22, 15], while relatedness research shows how actors enter activities near their existing capabilities across products, industries, technologies, and research fields [19, 28, 6, 16]. Soft-law research distinguishes flexible, non-binding responses from instruments with stronger legal obligations [1]. This distinction can turn on a single word. Immediately before adoption of the Paris Agreement, “shall” became “should” in Article 4.4, leaving procedural duties binding while making the expectation of economy-wide emission-reduction targets recommendatory [5, 30]. Our analysis cannot reconstruct comparable bargaining within the ATS, but it shows why formal instruments should not be treated as equivalent outcomes. The document records from the Antarctic Treaty System bring these patterns into one view, allowing us to follow documentary attention from actor portfolios, through the collective agenda, to the categories that Resolutions address.

Formal continuity appears most clearly within the Measure record, where later Measures cite 173 distinct

earlier Measures but only eight Decisions and nine Resolutions. Resolution-to-Measure citations also concentrate in Area protection and management (fig. S10). The liability case shows why this record remains incomplete: its final Measure omits intermediate instruments from its citations (section B.3). These citations establish formal lineage, while the concentration of Measures in Antarctic Specially Protected Area and management-plan work limits what they can show about movement from soft law to binding law.

Submitted papers record documentary effort, while preferences, bargaining, power, and informal rules shape which problems enter the record and how participants agree on formal texts [3, 33, 12]. Institutional output also differs from implementation and environmental change [18, 10]. The concern space measures overlap among actor portfolios, and the paper-output comparison depends on an author-coded category mapping. An independent blinded AI coder agreed with 38 of 45 assignments, and section B.2 examines the seven disagreements. Within these boundaries, documentary effort shows how participants organize attention before agreement.

Formal texts record the agreement reached by participants, while the space of concerns records where they directed attention beforehand.

Author Contributions

Casper van Elteren: conceptualization; methodology; software; formal analysis; investigation; data curation; visualization; writing, original draft; writing, review and editing. **Fiona Lippert:** conceptualization; writing, review and editing. **Vítor V. Vasconcelos:** conceptualization; writing, review and editing. **Zach Carter:** writing, review and editing. **Maria Kleshnina:** supervision; funding acquisition; writing, review and editing. **Michael Bode:** writing, review and editing. All authors contributed to the final manuscript.

Declaration of interests

All authors declare no competing interests.

Declaration of use of AI

The manuscript was edited by large language models for grammar (GPT-5.6 Sol), narrative flow and to ensure that the numbers in the manuscript match the outcome from our provided analysis scripts.

Ethics declaration

This study analyses publicly available institutional documents and does not involve human participants, personal data, or animal research.

Resource availability

Lead contact: Casper van Elteren (casper.vanelteren@qut.edu.au).

Materials availability: This study analyses existing documentary records and generated no new materials.

Data and code availability: The June 2026 analysis archive is available at Zenodo (<https://doi.org/10.5281/zenodo.20821775>). We have prepared a self-contained review archive with the reconstructed many-to-many paper panel, the pinned corpus of 277 official Measure body texts, formal-output inputs, analysis scripts, and derived tables. We will deposit the same materials as a new version of the Zenodo record before publication. Analysis scripts use Python 3.13 with NumPy, pandas, and SciPy; figures use UltraPlot [13]. We maintain development code at <https://github.com/cvanelteren/theSpaceOfConcerns>.

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References

- [1] Kenneth W. Abbott and Duncan Snidal. "Hard and Soft Law in International Governance". In: *International Organization* 54.3 (2000), pp. 421–456. doi: 10.1162/002081800551280.
- [2] Antarctic Treaty Secretariat. *Area Protection and Management / Monuments*. URL: <https://www.ats.aq/e/protected.html> (visited on 08/24/2026).
- [3] Peter Bachrach and Morton S. Baratz. "Two Faces of Power". In: *American Political Science Review* 56.4 (1962), pp. 947–952. doi: 10.2307/1952796.
- [4] Frank R. Baumgartner and Bryan D. Jones. *Agendas and Instability in American Politics*. Chicago: University of Chicago Press, 1993.
- [5] Daniel Bodansky. "The Legal Character of the Paris Agreement". In: *Review of European, Comparative & International Environmental Law* 25.2 (2016), pp. 142–150. doi: 10.1111/reel.12154.
- [6] Ron Boschma. "Relatedness as driver of regional diversification: a research agenda". In: *Regional Studies* 51.3 (2017), pp. 351–364. doi: 10.1080/00343404.2016.1254767.
- [7] British Antarctic Survey. *The Antarctic Treaty (1959)*. 2018. URL: <https://www.bas.ac.uk/about/antarctica/the-antarctic-treaty/the-antarctic-treaty-1959/> (visited on 01/29/2026).
- [8] Sarah Burch et al. "New Directions in Earth System Governance Research". In: *Earth System Governance* 1 (2019), p. 100006. doi: 10.1016/j.esg.2019.100006.
- [9] Zachary T. Carter et al. "Emerging Threats to Antarctic Conservation". In: *Nature Ecology & Evolution* 9.10 (Aug. 2025), pp. 1885–1896. doi: 10.1038/s41559-025-02814-4. (Visited on 04/23/2026).
- [10] David Coen et al. "Making Global Public Policy Work: A Survey of International Organization Effectiveness". In: *Global Policy* 13.5 (2022), pp. 656–668. doi: 10.1111/1758-5899.13125.
- [11] Klaus Dodds. "Governing Antarctica: Contemporary Challenges and the Enduring Myth of Exceptionalism". In: *The Polar Journal* 9.2 (2019), pp. 280–297. doi: 10.1080/2154896X.2019.1641543.
- [12] Julia Drubel and Janne Mende. "The hidden contestation of norms: Decent work in the International Labour Organization and the United Nations". In: *Global Constitutionalism* 12.2 (2023), pp. 246–268. doi: 10.1017/S2045381722000259.
- [13] Casper van Elteren and Matthew R. Becker. *UltraPlot: A succinct wrapper for Matplotlib*. Version 1.57.1. 2025. URL: <https://github.com/Ultraplot/Ultraplot>.
- [14] Baylor Fox-Kemper et al. "Ocean, Cryosphere and Sea Level Change". In: *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Ed. by Valérie Masson-Delmotte et al. Cambridge, United Kingdom and New York, NY, USA: Cambridge University Press, 2021, pp. 1211–1362. doi: 10.1017/9781009157896.011.
- [15] Christoffer Green-Pedersen and Stefaan Walgrave, eds. *Agenda Setting, Policies, and Political Systems: A Comparative Approach*. Chicago: University of Chicago Press, 2014.
- [16] Miguel R. Guevara et al. "The research space: using career paths to predict the evolution of the research output of individuals, institutions, and nations". In: *Scientometrics* 109.3 (2016), pp. 1695–1709. doi: 10.1007/s11192-016-2125-9.
- [17] António Guterres. *Secretary-General's Video Message from Antarctica*. United Nations. Nov. 2023. URL: <https://www.un.org/sg/en/content/sg/statements/2023-11-25/secretary-generals-video-message-antarctica> (visited on 08/25/2026).
- [18] Tamar Gutner and Alexander Thompson. "The Politics of IO Performance: A Framework". In: *The Review of International Organizations* 5.3 (2010), pp. 227–248. doi: 10.1007/s11558-010-9096-z.
- [19] C. A. Hidalgo et al. "The Product Space Conditions the Development of Nations". In: *Science* 317.5837 (July 2007), pp. 482–487. ISSN: 0036-8075. doi: 10.1126/science.1144581. (Visited on 12/13/2024).
- [20] César A. Hidalgo et al. "The Principle of Relatedness". In: *Unifying Themes in Complex Systems IX*. Ed. by Alfredo J. Morales et al. Cham: Springer, 2018, pp. 451–457.

- [21] Paul R. Holland et al. "The Southern Ocean and its interaction with the Antarctic Ice Sheet". In: *Science* 367.6484 (Mar. 2020), pp. 1326–1330. DOI: 10.1126/science.aaz5491.
- [22] Bryan D. Jones and Frank R. Baumgartner. *The Politics of Attention: How Government Prioritizes Problems*. Chicago: University of Chicago Press, 2005.
- [23] Christopher C. Joyner. *Governing the Frozen Commons: The Antarctic Regime and Environmental Protection*. Columbia, SC: University of South Carolina Press, 1998.
- [24] II Kennicutt Mahlon C. et al. "Sustained Antarctic Research: A 21st Century Imperative". In: *One Earth* 1.1 (Sept. 2019), pp. 95–113. ISSN: 2590-3322. DOI: 10.1016/j.oneear.2019.08.014.
- [25] Rakhyun E. Kim. "Is Global Governance Fragmented, Polycentric, or Complex? The State of the Art of the Network Approach". In: *International Studies Review* 22.4 (2020), pp. 903–931. DOI: 10.1093/isr/viz052.
- [26] Rakhyun E. Kim. "The Emergent Network Structure of the Multilateral Environmental Agreement System". In: *Global Environmental Change* 23.5 (2013), pp. 980–991. DOI: 10.1016/j.gloenvcha.2013.07.006.
- [27] Ming Luo et al. "Regulate to protect fragile Antarctic ecosystems from growing tourism". In: *Nature* 631 (2024), p. 275. DOI: 10.1038/d41586-024-02250-7.
- [28] Frank Neffke, Martin Henning, and Ron Boschma. "How Do Regions Diversify over Time? Industry Relatedness and the Development of New Growth Paths in Regions". In: *Economic Geography* 87.3 (2011), pp. 237–265. DOI: 10.1111/j.1944-8287.2011.01121.x.
- [29] Mark Poynting. *World's Oceans Hit Highest Temperature on Record as El Niño Grows*. BBC News. Aug. 2026. URL: <https://www.bbc.com/news/articles/c62m4ggnp78o> (visited on 08/25/2026).
- [30] Lavanya Rajamani. "The 2015 Paris Agreement: Interplay Between Hard, Soft and Non-Obligations". In: *Journal of Environmental Law* 28.2 (2016), pp. 337–358. DOI: 10.1093/jel/eqw015.
- [31] Daniela P. Sampaio. "Diplomatic culture and institutional design: Analyzing sixty years of Antarctic Treaty governance". In: *Anais da Academia Brasileira de Ciências* 94 (suppl 1 2022), e20210539. ISSN: 1678-2690, 0001-3765. DOI: 10.1590/0001-3765202220210539. URL: http://www.scielo.br/scielo.php?script=sci_arttext&pid=S0001-37652022000201803&tlng=en (visited on 01/08/2026).
- [32] Richard H. Steinberg. "In the Shadow of Law or Power? Consensus-Based Bargaining and Outcomes in the GATT/WTO". In: *International Organization* 56.2 (2002), pp. 339–374. DOI: 10.1162/002081802320005504.
- [33] Randall W. Stone. "Informal Governance in International Organizations: Introduction to the Special Issue". In: *The Review of International Organizations* 8.2 (June 2013), pp. 121–136. ISSN: 1559-744X. DOI: 10.1007/s11558-013-9168-y. (Visited on 06/10/2025).
- [34] V. A. Traag, L. Waltman, and N. J. van Eck. "From Louvain to Leiden: Guaranteeing Well-Connected Communities". In: *Scientific Reports* 9.1 (Mar. 2019), p. 5233. ISSN: 2045-2322. DOI: 10.1038/s41598-019-41695-z. (Visited on 06/17/2025).
- [35] Joanne Yao. "An International Hierarchy of Science: Conquest, Cooperation, and the 1959 Antarctic Treaty System". In: *European Journal of International Relations* 28.4 (2022), pp. 939–963. DOI: 10.1177/13540661211033889.
- [36] Parsa Yaryab. *Parsayarya/Scraping-ATCM: ATCM WP and IP paper dataset*. Version V1.0. Zenodo, Jan. 2024. DOI: 10.5281/zenodo.10578846. URL: <https://doi.org/10.5281/zenodo.10578846>.

Supplementary Information

The Supplementary Information follows the order of the main claims and retains only analyses needed to define, support, or bound them.

- **Data and measurement.** Archive construction, the multi-label institutional taxonomy, actor types, the concern space, a count-reconstruction test, and output positioning.
- **Portfolio development.** Tests of specialization shifts, duration, cohort differences, and robustness.
- **From attention to output.** Repeated predictions trained on earlier meetings for each form of output, an audit of formal citations, aggregated association checks, and the Madrid Protocol boundary test.

A Data and Methods

A.1 Data and archive coverage

We analyse working papers and information documents submitted from 1961 to 2025. “ATS” refers to the broader governance system anchored in the Antarctic Treaty, but our corpus comes only from the Treaty-centred archive of Antarctic Treaty Consultative Meetings (ATCMs). The Secretariat’s taxonomy also includes topics handled by the Committee for Environmental Protection because these matters reach the ATCM agenda. Meeting volume rises partly for institutional reasons: ATCMs became annual in the 1990s, and the Committee has normally met annually since 1998. Conference Room Papers lack systematic availability, so we exclude them. The corpus therefore observes formal documentary preparation, not the informal negotiation record.

The official archive supplies thematic search categories, submitters, and document types [36]. We retain Secretariat papers, so the Secretariat appears both as maintainer of the database categories and as one of the 66 submitting actors. Its papers enter on the same terms as those of other submitters.

We collected the data on 6 February 2026 from the ATS website (<https://www.ats.aq/devAS/Meetings/DocDatabase?lang=e>) with Carlo Hamalainen’s antarctic-database-go collector (<https://github.com/carlohamalainen/antarctic-database-go>). The collector stores HTTP responses, source documents, and processing status in SQLite. These cached records permit deterministic reruns even if the website changes.

We do not derive the categories from paper text. They are the harmonized cross-meeting categories under which the Secretariat-maintained database returns each paper. A paper can appear under several categories, although the returned paper payload does not itself carry a category field. The query responses also record no category-assignment date. The concern space therefore describes shared specialization within an institutional taxonomy, not semantic similarity. We will include the processed many-to-many panel and reconstruction code in the next Zenodo version.

We reconstructed paper–category relations from the collector’s cached category-query responses, then deduplicated attachment-linked records by `paper_id`. This produced 6,591 unique papers. We removed 18 papers (0.3%, all after 2010) returned only under the

placeholders ALL or Other, leaving 6,573 papers and 7,895 substantive paper–category relations. Of the retained papers, 5,345 have one category and 1,228 have two to five. We divide a paper with k categories equally among them, so it contributes $1/k$ to each label. Papers with multiple sponsors are then expanded to one row per submitting actor; each sponsor receives the paper’s same fractional category weights. The extracted and reconstructed dataset contains the following columns.

A separate snapshot from 29 January 2024 contains the same 45 categories. All 6,230 matched papers with a non-missing snapshot category retain that category in the 2026 reconstruction. The snapshot stores only one category per paper, does not date that assignment, and does not cover ATCMs 46–47. It supports label stability but cannot establish when every multi-label membership became available.

<code>meeting_year</code>	<code>paper_id</code>
<code>meeting_type</code>	<code>party_type</code>
<code>meeting_number</code>	<code>paper_name</code>
<code>meeting_name</code>	<code>paper_number</code>
<code>party</code>	<code>paper_revision</code>
<code>category</code>	<code>paper_language</code>
<code>page_url</code>	<code>paper_url</code>
<code>page_nr</code>	<code>exists</code>
<code>payload_json</code>	<code>agendas</code>
<code>attachment_id</code>	<code>parties</code>
<code>attachment_name</code>	<code>attachment_language</code>
<code>attachment_url</code>	<code>attachment_exists</code>

The June 2026 archive is available at <https://doi.org/10.5281/zenodo.20821775>; we will add the reconstructed many-to-many panel in the next version. Figure S1a summarizes the resulting distribution across concerns, while panel b shows annual submission volume.

The unit is the *submitting actor*, not legal status under the Treaty. We include Consultative Parties, Non-Consultative Parties, Observers, invited Expert organizations, and the Secretariat because all appear as submitters in this documentary arena. Only Consultative Parties take decisions. The actor types are therefore not institutionally equivalent, and aggregated estimates describe documentary activity rather than formal authority [31]. Sixty-six actors appear in the corpus. The extracted field `party_type` contains Working Paper and Information Paper codes, not Treaty status. Actor-role information is separate and can change over time. The cohort comparison retains 43 actors, while the states-only locality check retains 42 states. Conference Room Papers do not enter any analysis.

A.2 Building the concern map

Counts and shared specialization answer different questions. Counts show where attention is common. Shared specialization shows which concerns receive a disproportionate share of the same actors’ papers. A concern pursued by nearly everyone can be prominent in counts but uninformative for shared specialization because no actor stands out. A small technical group can produce the opposite pattern. Our map records which concerns share specialized actors, not the overall importance of a concern.

Let $x(a,i)$ denote the fractionally counted submissions by actor a on concern i . If the archive returns paper p under k_p substantive categories, it contributes $1/k_p$ to each; a co-sponsored paper contributes those weights to every listed submitting actor. Actors include Consultative Parties, Non-Consultative Parties, Observers, invited Expert organizations, and the ATS Secretariat whenever

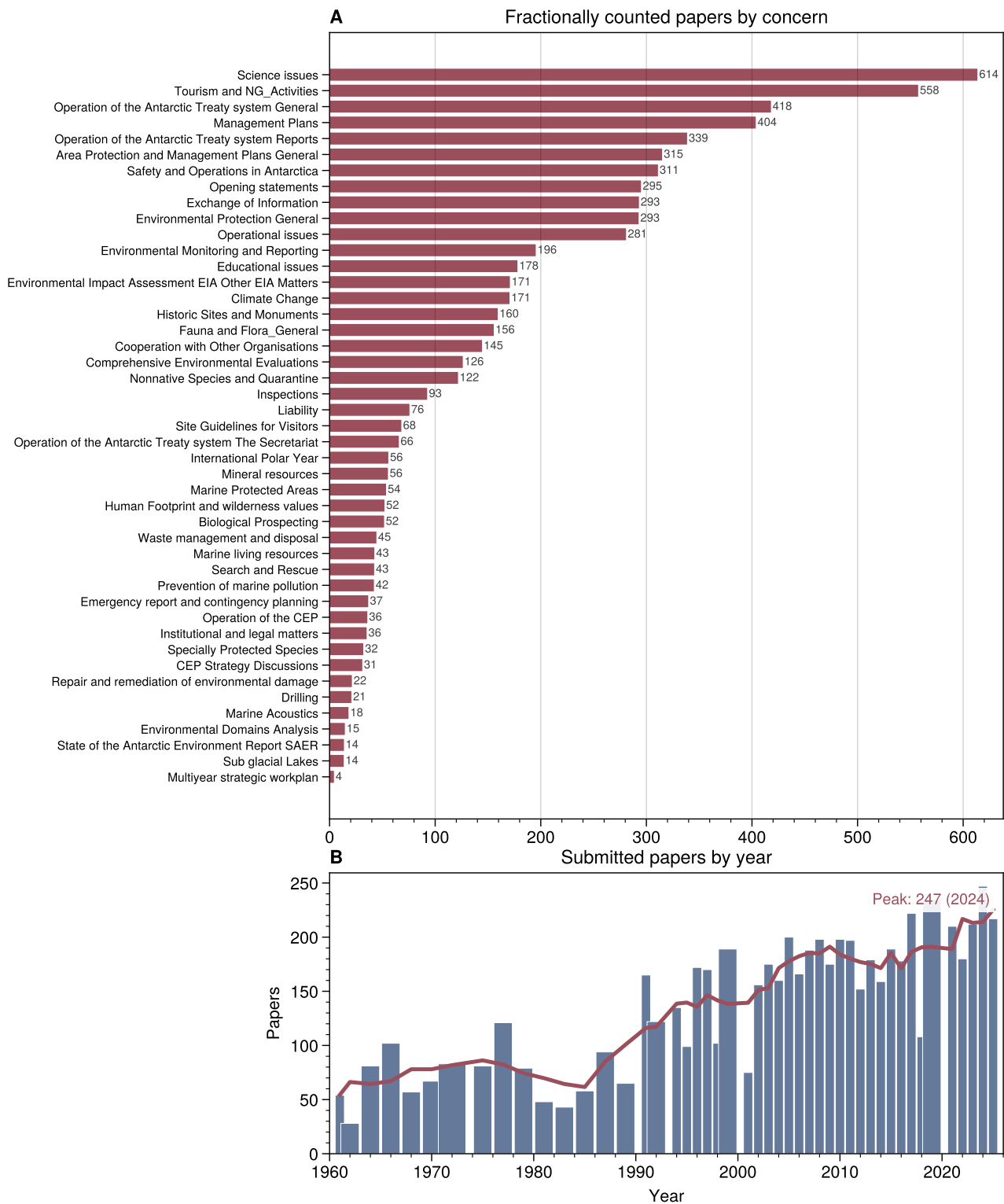


Figure S1: Paper submissions rise after 1990 and remain uneven across concerns. (a) Fractional counts for 6,573 working and information papers by Secretariat concern, 1961–2025. A paper returned under k concerns contributes $1/k$ to each. (b) Unique papers submitted each year. The line is the centred five-year average.

they appear as submitters. We define the normalised activity share

$$\pi(a, i) = \frac{x(a, i)}{\sum_j x(a, j)}, \quad (1)$$

and the system-wide importance of topic i as

$$Z_i = \frac{\sum_c x(c, i)}{\sum_{d, j} x(d, j)}. \quad (2)$$

The Revealed Policy Advantage (RPA) index is then

$$\rho(a, i) = \frac{\pi(a, i)}{Z_i}. \quad (3)$$

We call this ratio *revealed policy advantage* rather than the usual *revealed comparative advantage* (RCA) because the units are policy concerns rather than exported products. The mathematics does not change, but the interpretation is narrower. Submissions have not passed a competitive market test. Actors may submit to inform, comply, signal, or seek action. RPA records the concerns receiving a disproportionate share of an actor's papers. It cannot separate intent from mandate, expertise, reporting duties, or document-production capacity, and it cannot observe priorities that produce no paper. Scientific capacity may also enter the measure because Article IX(2) ties consultative status to the scale of scientific research activity. Nor do concerns have an equivalent of product complexity or economic value. Shared specialization remains useful because it records how actors allocate finite documentary effort across the agenda.

We measure how strongly each pair of concerns shares specialized actors. We call this value proximity:

$$\phi(i, j) = \min \left\{ \begin{array}{l} \Pr(\rho(a, i) \geq 1 \mid \rho(a, j) \geq 1), \\ \Pr(\rho(a, j) \geq 1 \mid \rho(a, i) \geq 1) \end{array} \right\}. \quad (4)$$

with

$$\Pr(\rho(a, j) \geq 1 \mid \rho(a, i) \geq 1) = \frac{\sum_a \mathbf{1}[\rho(a, i) \geq 1 \wedge \rho(a, j) \geq 1]}{\sum_a \mathbf{1}[\rho(a, i) \geq 1]}, \quad (5)$$

This produces a weighted map of relations among concerns. Conditional probability allows two uncommon concerns to be close when specialization in one reliably accompanies specialization in the other; a joint probability would shrink mechanically with rarity. Taking the lower of the two directions prevents a niche concern from appearing close to a broad concern merely because the niche usually occurs alongside it. The resulting score is symmetric. We estimate it across actors in the relevant window. With 66 actors, individual relations rest on modest counts: the strongest, between Marine living resources and Mineral resources, rests on 6 actors. The map is therefore a coarse aggregate rather than a high-resolution estimate of every pair. We construct the full-record map and rebuild time-specific maps within each window. This adapts product-space proximity to governance concerns [19].

A.3 Can marginal counts reconstruct the concern space?

To test whether proximity is only a reformulation of prevalence, we treat each of the 990 concern pairs as one observation and use full-record proximity $\phi(i, j)$ as the outcome. For each pair, the paper-count model contains the sum, absolute difference, and minimum of the two concerns' log-transformed marginal paper volumes. The actor-prevalence model uses the corresponding functions of the number of actors with $\text{RPA} \geq 1$. A combined model includes both sets. These are reconstruction diagnostics, not causal models.

We assess generalization with a leave-endpoint-out procedure. For each concern, we omit every pair containing it, fit the model to the remaining pairs, and predict the omitted relations. We predict each pair twice, using one fit that excludes each endpoint,

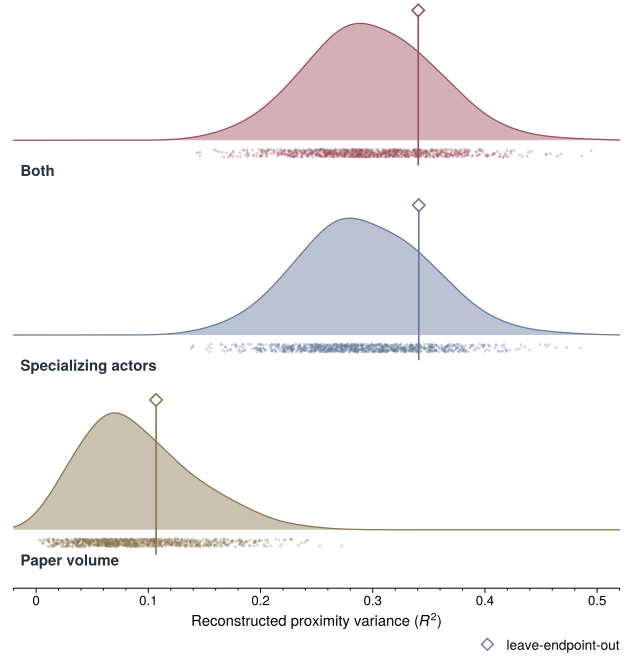


Figure S2: Marginal counts reconstruct only part of shared specialization. Half-violins and rugs show pooled R^2 across 1,000 actor bootstraps and 990 concern pairs for models using paper volume, specializing-actor counts, or both. Open diamonds mark leave-endpoint-out R^2 , which predicts pairs containing a concern excluded from fitting.

and average the two predictions before calculating R^2 . We then resample the 66 complete actor histories 1,000 times and rebuild paper volume, RPA assignments, the full shared-specialization matrix, and every model in each draw. Finally, 9,999 quadratic assignment procedure (QAP) label permutations randomly reassign the observed marginal attributes to concerns while leaving the matrix fixed. This preserves the marginal distributions while testing whether their observed alignment with the concern map exceeds arbitrary assignments.

Paper volume explains 12.8% of pairwise shared-specialization variation in the pooled fit and 10.7% when predicting relations involving a concern omitted from model fitting (fig. S2 and table S1). Specialization prevalence explains 35.3% under that fit and 34.1% under the same test. The combined model reaches 35.9% across all pairs but remains at 34.1% when the procedure omits concerns. Across actor bootstraps, the additional all-pairs variance explained by paper volume after actor prevalence has a median of 0.005 and a central 95% interval of [0.0003, 0.0173]. All three observed pooled R^2 values exceed every corresponding QAP draw ($p = 0.0001$). Marginal prevalence therefore aligns with the concern space, as expected for a score constructed from specialization, but it does not reconstruct most pairwise variation. `scripts/counts_vs_relations.py` reproduces the analysis.

A.4 Testing whether specialization shifts toward nearby concerns

We compare adjacent rolling five-meeting windows. A concern enters the comparison only when the actor was not already specialized in it. For actor a , concern i , and transition from $t-1$ to t , this condition is

$$\text{at-risk}_{ait} = 1 \iff \rho_{ai, t-1} < 1.$$

The binary outcome is then

$$y_{ait} = 1 \iff \rho_{ai, t} \geq 1 \text{ and } \rho_{ai, t-1} < 1.$$

Table S1: Marginal counts recover only part of pairwise shared specialization. Pooled R^2 uses all 990 concern pairs. Leave-endpoint-out R^2 predicts pairs containing an excluded concern. Actor-bootstrap entries are medians with central 95% intervals from 1,000 resamples. Quadratic assignment procedure (QAP) p values use 9,999 random assignments of the marginal attributes.

Model	Pooled R^2	Leave-endpoint-out R^2	Actor-bootstrap R^2	QAP p
Paper volume	0.128	0.107	0.083 [0.016,0.199]	0.0001
Specializing actors	0.353	0.341	0.288 [0.182,0.409]	0.0001
Both	0.359	0.341	0.295 [0.188,0.413]	0.0001

Let the previously active support be

$$S_{a,t-1} = \{j : \rho_{aj,t-1} \geq 1\},$$

let A_{t-1} be the concerns that have appeared in the archive before the transition, and let the available set be

$$R_{at} = \{i \in A_{t-1} : \rho_{ai,t-1} < 1\},$$

and let $m_{at} = \sum_{i \in R_{at}} y_{ait}$ denote the number of specialization crossings in an actor-period. Our main locality specification is then an actor-period conditional logit,

$$\Pr(\mathbf{y}_{at} | m_{at}, X_{at}) \propto \exp \left(\sum_{i \in R_{at}} y_{ait} (\beta_d d_{ai,t-1} + \beta_p \text{pop}_{i,t-1}) \right), \quad (6)$$

where

$$d_{ai,t-1} = 1 - \max_{j \in S_{a,t-1}} \phi(i, j)$$

is the distance from candidate concern i to the actor's earlier portfolio and $\text{pop}_{i,t-1}$ is its prior popularity (the share of actors with $\text{RPA} \geq 1$). The model holds constant how many concerns cross the threshold in that period. It therefore compares which available concerns cross for the same actor at the same time, rather than comparing actors with different general rates of expansion. Periods in which an actor has no earlier portfolio or no threshold crossing do not enter the model. Differences in the number of concerns already held are constant within each actor-period comparison. The map, portfolio membership, and crossings all use $\text{RPA} \geq 1$.

These restrictions leave 968 actor-period comparisons and 30,957 actor-period-comparisons. The distance coefficient is -2.02 (standard error [SE] 0.22), or an odds ratio of 0.817 per +0.1 distance (95% confidence interval [CI] [0.783, 0.853]). The specification is the primary cause neither the map nor the choice set uses information from the first observed transition. A second outcome requires no paper on the concern in the earlier window and at least one in the next. It gives an odds ratio of 0.874 ([0.834, 0.916]), showing that the local association is not only a consequence of the relative specialization threshold. `scripts/compare_major_results_category.py` reproduces both estimates.

We construct the example trajectories in Supplementary fig. S3 on a continuous axis. We compute RPA in rolling five-meeting windows and locate each actor on the horizontal axis of the fixed full-record map as the RPA-weighted mean of its concern coordinates.

We omit windows with fewer than three concerns at $\text{RPA} \geq 1$. Australia connects this continuous summary to the snapshots in fig. 2, while the Netherlands and Ukraine provide two further examples. The trajectories illustrate movement rather than testing locality. `figS22_actor_trajectories.py` reproduces the figure.

Because the rolling windows overlap, successive choice sets for the same actor are not statistically independent. We therefore resample complete actor histories and rebuild portfolios, historical availability, the map from all earlier records, and the choice model over 500 bootstrap replicates. We also use disjoint five-meeting blocks and restrict the panel to states. The blocks reduce rolling-window overlap, but successive actor transitions still share their middle block and use maps built from nested sets of earlier records; the actor-history bootstrap addresses this dependence. The association remains negative in both checks; with disjoint blocks, the odds ratio is 0.904 ([0.855, 0.957]). We report these checks in table S6. The result also remains stable across the full-record, previous-window, and earlier-record maps while holding fixed the set of options available at the time (fig. S5). The primary distance uses raw $1 - \phi$, which has bounds and a direct interpretation; $-\log \phi$ is unstable when two concerns have zero proximity. `scripts/bootstrap_fractional_locality.py` and `scripts/hazard_robustness_dependence.py` reproduce the dependence checks.

A.5 How long specialization lasts

We estimate specialization spells on 1-year windows. A spell begins when an actor first crosses $\text{RPA} \geq 1$ for a concern, and we follow it until inactivity exceeds an allowed gap of one to five consecutive years. We right-censor spells still active at the end of 2025 and compute a Kaplan–Meier curve for each gap definition. The allowed gap delays any failure declaration, so every curve mechanically equals one over that initial interval. Figure S4 reports all five definitions on the unshifted time-since-entry scale. The analysis does not distinguish a concern-specific lapse from years in which the actor submits nothing, so it is a descriptive sensitivity analysis rather than evidence of a retention mechanism.

A.6 A retrospective comparison at the same career stage

The cohort analysis in section B.6 compares the concerns in each actor's opening portfolio with an opening-window benchmark. The benchmark weights concerns by their share of recorded specializations during that actor's first 15 years. It scores holder breadth and nearby relations using the full-record map, so this is a retrospective description rather than a reconstruction of what actors could know at the time.

An actor is eligible if at least $\text{WIN} = 15$ years separate its first and last appearance in the archive; the requirement excludes members joining after 2010 by construction, since their record cannot yet span the window. Its *opening window* runs from its first recorded year to that year plus $\text{WIN} - 1$. Within that window we rebuild the actor–concern interaction matrix from the submissions falling inside it, compute RPA exactly as in the full-record construction, and treat a concern as held when $\text{RPA} \geq 1$. We drop actors holding fewer than three specialized concerns in their own opening window to avoid estimating

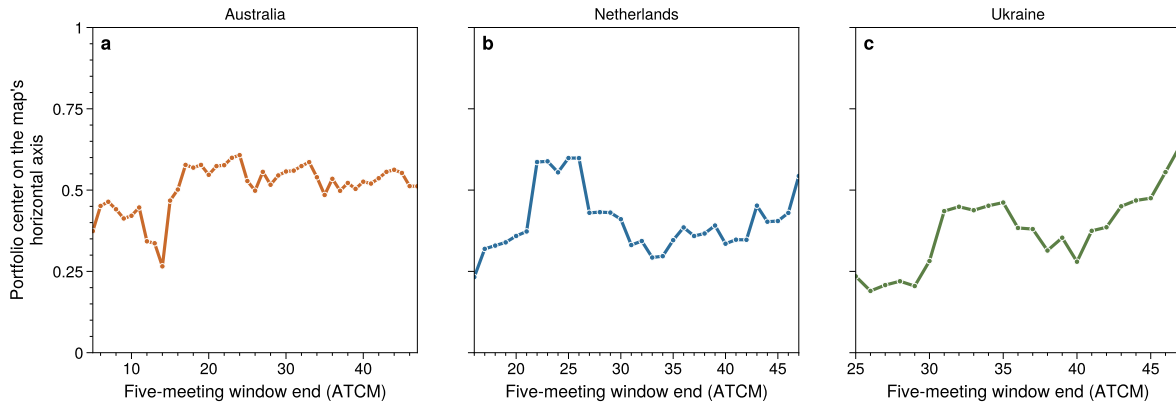


Figure S3: Three actor portfolios move across the concern map. Lines trace the revealed-policy-advantage-weighted mean horizontal coordinate of Australia, the Netherlands, and Ukraine on the fixed full-record map over rolling five-meeting windows. We omit windows with fewer than three specialized concerns. The trajectories illustrate movement only.

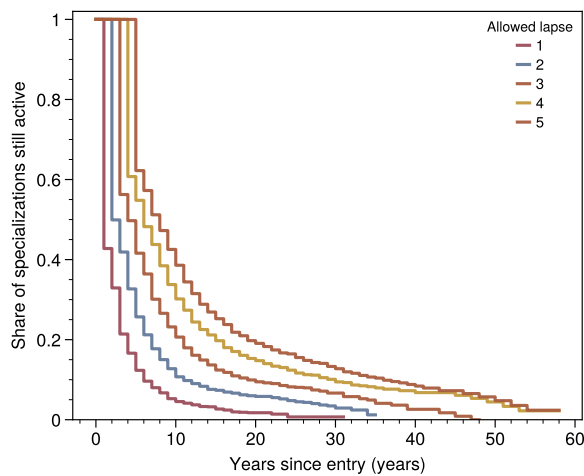


Figure S4: Estimated specialization duration depends on the allowed inactivity gap. Kaplan–Meier curves show the share of annual specialization spells, defined by revealed policy advantage (RPA) ≥ 1 , still active when failure requires more than one to five consecutive inactive years. The five-year curve remains at one through year five. Survival after ten years ranges from 0.05 to 0.42. After twenty years every estimate is below 0.20. We right-censor spells active in 2025. The archive cannot distinguish a concern-specific lapse from an actor-year with no submissions.

opening position from one or two concerns. This leaves 14 actors joining 1961–80, 12 joining 1981–90, and 17 joining 1991–2010.

The comparison asks whether the actors specializing in those concerns hold broader or narrower portfolios across the full record than expected, and whether those concerns have more or fewer nearby neighbours in the full-record map. The opening-window benchmark weights concerns by their share of recorded specialization during each actor’s opening window. The 1991–2010 cohort sits at 0.929 of this benchmark for specialized actors’ full-record portfolio size (95% actor-bootstrap interval [0.889,0.966]) and 0.883 for neighbours in the full-record map ([0.798,0.974]). The two earlier cohorts sit near parity on both measures. Each of 400 bootstrap replicates resamples the 66 actors and rebuilds opening portfolios and availability weights while retaining the fixed concern scores. `scripts/cohort_partition_free.py` reproduces the comparison.

A.7 Checks on local specialization

This subsection tests whether local specialization depends on the temporal construction of the space, the threshold-crossing definition, self-contribution to the shared-specialization matrix, or full rather than fractional credit for co-sponsored papers. A separate subsection below asks whether a small number of actors dominates the recovered space.

We first change how we build the concern map while holding fixed the set of options available at the time. Figure S5 compares the full-record, previous-window, and earlier-record maps. In each case, distance is $d = 1 - \max \phi$. Moving 0.1 farther gives odds ratios of 0.67, 0.78, and 0.82, respectively. Each model adjusts for prior concern popularity when estimating the distance association. The earlier-record map is primary because it uses no later shared-specialization information. We retain raw $1 - \phi$ because its values lie between zero and one; taking its logarithm is unstable when two concerns have a score of zero.

One check changes the definition of the map. An actor can cross the RPA threshold because a concern becomes less common even when its own paper count does not rise. We therefore require that the actor concern filed in the paper earlier window and at the least one in the next. In the corresponding actor-period model, moving 0.1 farther gives an odds ratio of 0.874 ([0.834, 0.916]), close to the primary direction. `scripts/compare_major_results_category_treatments.py` reproduces the check.

Because each actor’s submission history contributes to the scores used to compute its distances, the locality test could contain a self-contribution. We therefore rebuild each actor’s map from all earlier records without that actor while holding the temporally ordered outcome panel fixed. We also divide each paper equally among its sponsors, in addition to dividing it among its categories. Both checks preserve the negative distance association. The local result based only on earlier information therefore does not depend on an actor contributing to its own map or on full co-sponsor credit. `scripts/hazard_circularity_checks.py` reproduces both analyses.

Table S2: Retrospective first-15-year concern positions differ across entrant cohorts. Ratios compare opening portfolios with an opening-window benchmark weighted by recorded specializations during the same first-15-year period. A value of 1 marks parity with that benchmark. Holder breadth is the mean full-record portfolio size of actors specializing in each concern. Nearby concerns are neighbours in the full-record map with $\phi \geq 0.30$. Intervals are actor-bootstrap 95% intervals from 400 resamples. n counts actors with a 15-year record and at least three opening-window specializations.

Cohort	Scale	Ratio	95% CI	n
1961–80	Holders' average portfolio size	1.00	[0.978, 1.016]	14
1961–80	Nearby concerns	0.99	[0.962, 1.021]	14
1981–90	Holders' average portfolio size	0.98	[0.952, 1.012]	12
1981–90	Nearby concerns	1.04	[0.975, 1.133]	12
1991–2010	Holders' average portfolio size	0.93	[0.889, 0.966]	17
1991–2010	Nearby concerns	0.88	[0.798, 0.974]	17

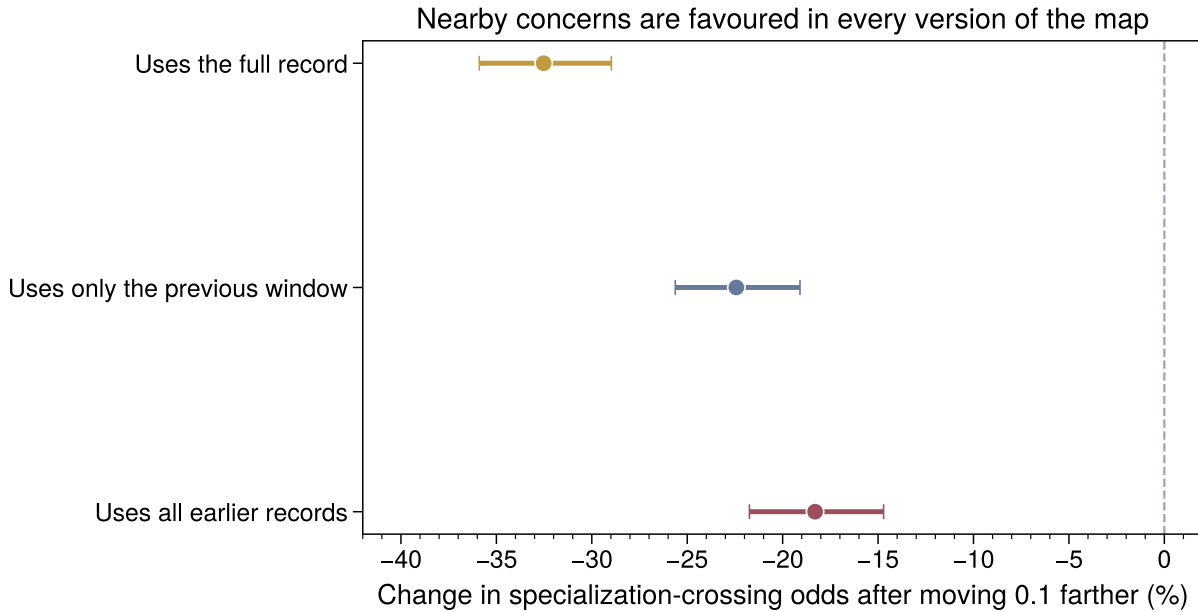


Figure S5: Local specialization persists across three temporal map definitions. Across 968 actor-periods, odds ratios give the change in odds of crossing $RPA \geq 1$ for a 0.1 increase in distance $d = 1 - \max \phi$ from the earlier portfolio. Values below one indicate locality. Bars are model-based 95% intervals. Maps use the full record, the previous five-meeting window, or all earlier records. The earlier-record map is primary because it uses only information available before each transition. Models compare options available to the same actor at the same time and adjust for prior popularity.

Although we compute no relation from text, the recovered map could still track the wording of the official labels. We compare the shared-specialization matrix with term frequency-inverse document frequency (TF-IDF) cosine similarity over the same 45 labels. We use word (1–2 gram, English stop words removed) and character (3–5 gram, word-boundary) analysers. A Mantel permutation with 10,000 joint row-and-column shuffles preserves dependence among the 990 pairwise entries. Lexical similarity is positively but weakly associated with shared specialization, explaining 0.9% of its variance for word n -grams ($r = 0.094$, Spearman $\rho = 0.066$, $p = 0.008$) and 1.5% for character n -grams ($r = 0.122$, Spearman $\rho = 0.157$, $p = 0.002$). This test does not remove concepts already built into the taxonomy; it shows only that literal label overlap explains little of the recovered relation. The full matrix remains dense but uneven, with stronger relations concentrated within parts of the agenda (fig. S6). `scripts/semantic_baseline.py` reproduces the numbers and pair table.

A.8 Reading the map at two levels of detail

We use the Constant Potts Model (CPM) to group concerns with relatively strong shared-specialization scores [34]. CPM receives ϕ directly as the strength of a relation, not $1 - \phi$ as a distance. The setting γ controls the level of detail. CPM tends to place two concerns in the same region when their score exceeds γ and separate

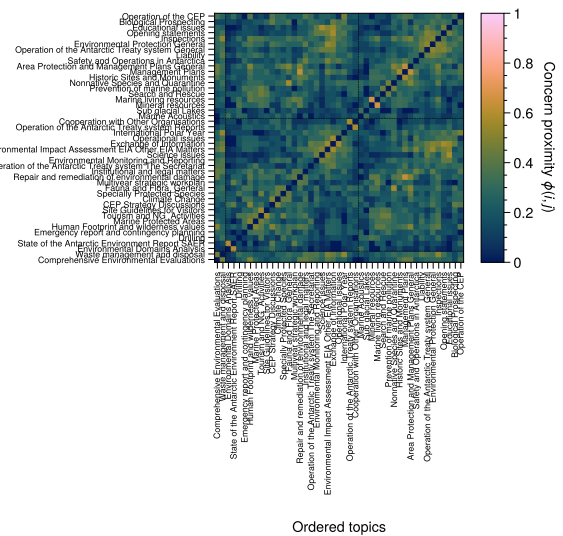


Figure S6: Shared specialization is widespread but uneven across 45 concerns. Each cell shows the full-record score across 66 actors, defined as the lower conditional probability of specialization in one concern given specialization in the other. Darker cells indicate a stronger relation. The diagonal equals one.

them when it falls below γ . Varying γ therefore gives progressively finer readings rather than one uniquely correct grouping.

The four-region reading at $\gamma = 0.211$ contains groups of 30, 7, 5, and 3 concerns. The seven-region reading at $\gamma = 0.273$ contains groups of 16, 13, 6, 4, 2, 2, and 2. The finer reading is mostly nested within the coarser one: 43 of 45 concerns fall within a single coarse parent, giving fine-to-coarse purity of 0.956. Two standard measures of agreement between groupings give an adjusted Rand index of 0.478 and normalised mutual information of 0.707. The exceptions are Cooperation with Other Organisations and ATS reports, which form one fine pair across two coarse regions, and Marine Acoustics and Sub-glacial Lakes, which form another. Figure S7 places both groupings on identical coordinates. `fig_cpm_partition_comparison.py` reproduces the comparison.

To assess uncertainty, we resample submitting actors 300 times. Within every sample, we recompute RPA, rebuild the full matrix, and refit CPM at $\gamma = 0.273$. We record how often each pair enters the same region across resamples, a quantity called pairwise co-assignment frequency. This avoids matching arbitrary region numbers across bootstrap samples. The diagonal of fig. S8a shows how often concerns from the same displayed region remain together; off-diagonal cells show merging across displayed regions. Internal subdivision dominates the pattern. The 16-concern procedural and scientific region has a mean within-region co-assignment frequency of 0.67 but typically splits into three pieces. The 13-concern environmental-management region is less stable, with a mean of 0.41. The Cooperation–Reports pair and the Environmental Domains–State of the Antarctic Environment Report (SAER) pair remain together in 79% and 66% of samples. Marine Acoustics and Sub-glacial Lakes remain together in 40%. Whole-region persistence is lower because every within-region pair must remain together (fig. S8b). The smallest gaps between displayed and alternative assignments identify the least stable concern placements (fig. S8c). We use pairwise co-assignment frequency as the primary comparison because whole-region persistence decreases mechanically with region size. `scripts/analyze_cpm_partition_uncertainty.py` reproduces the bootstrap.

A.9 Do actor breadth and concern prevalence explain the map?

We test whether the number of concerns held by each actor and the number of actors specializing in each concern explain the map. The shuffled comparison begins with the binary actor–concern matrix at $RPA \geq 1$. Curveball trades randomize which actors hold which concerns while preserving both sets of totals. We generate 1,000 maps and recompute the shared-specialization scores, number of positive relations, overall connectivity, and weighted modularity after each draw. We optimize modularity separately in every observed and shuffled map using four independent starts of the Louvain algorithm. Empirical upper-tail probabilities include the observed statistic.

The seven displayed CPM re-observed come map, from the holding their labels fixed in shuffling draws evidence gives rather support-an independent test. A second comparison identifies the best grouping separately within every draw. We also repeat the shuffling after filtering actors at paper-volume thresholds of 5, 10, 20, and 50 and held-concern thresh-

olds of 3 and 5. A mixing check compares chains with 5, 20, and 50 trades per observed membership. `scripts/topology_fixed_margin_null.py` reproduces the comparison, and `output/scientific_checks/topology_curveball_mixing_summary.js` records the mixing check.

A.10 Does any one actor determine the map?

We test whether any one actor determines the map by removing actors before rebuilding it. At each step, we remove the actor whose exclusion produces the largest root mean squared difference from the original 66-actor matrix, then repeat the procedure. This does not test a substantive behavioral claim directly; it asks whether many actors jointly produce the pattern or a small number dominate it.

In fig. S9, the space changes gradually as we remove actors, indicating that no one submitter drives the pattern. The comparison becomes less informative near the end of the sequence, when too few actors remain to estimate the original relations. This test shows that the full-record map draws on many submitters; it does not establish that individual relation ranks or the map remain fixed over time.

A.11 Aligning papers and adopted texts by category

The output inventory contains all 584 records with titles from regular ATCMs 19–47 (1995–2025): 277 Measures, 135 Decisions, and 172 Resolutions. We begin in 1995 because Decision 1 (1995) established these three forms of output and the official register is complete for the period. Measures become legally binding after the required approvals. Decisions address internal organizational matters and are operative on adoption or another stated date. Resolutions are hortatory and not legally binding. We therefore group them only as *adopted ATCM outputs*; the count does not imply equal legal effect. Instruments adopted at Special ATCMs, including the 1991 Madrid Protocol, are outside this inventory. Each record retains instrument type, number, year, and meeting because numbers repeat across years. The analysis reads these records directly from the archived 9 February 2026 official export and verifies its SHA-256 checksum, total counts, type counts, and counts by meeting.

The Secretariat's paper archive and official instrument register use different classifications: 45 concerns for papers and 15 broad categories for adopted texts. Adopted texts do not enter the 45-concern paper map as additional concerns. We aggregate paper attention into the 15 official instrument categories. This alignment compares the aligned paper totals with the unchanged official category assignments of adopted instruments. It does not treat instruments as points in the concern space or use citations from instruments to papers. For a paper returned under several fine concerns, we sum its fractional weights within official instrument categories; they still total one. Sixty instruments have more than one official category, giving 646 instrument–category relations; an instrument with k categories contributes $1/k$ to each. The resulting weights also total one for every instrument.

Neither Secretariat source links the two classifications, so we construct a 45-to-15 alignment that assigns each paper concern to the official instrument category that best matches its institutional function. This alignment changes how we aggregate paper attention; it does not recode adopted instruments. For example, Management Plans, Environmental Domains Analysis, Marine Protected Areas, and the general area-protection concern all enter “Area protection

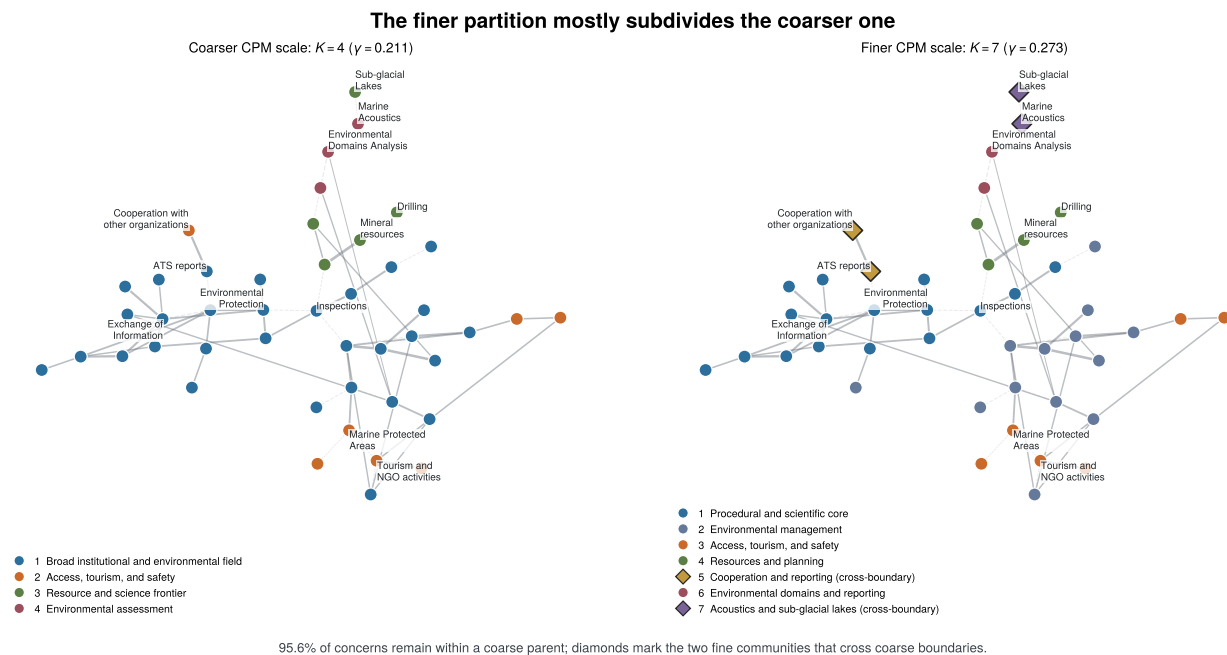


Figure S7: Seven Constant Potts Model regions mostly subdivide the four-region structure. The panels show $\gamma = 0.211$ ($K = 4$) and $\gamma = 0.273$ ($K = 7$) with identical concerns, relations, and coordinates. Related colours connect fine regions to coarse parents. Diamonds mark the two fine pairs that cross coarse boundaries. Overall, 95.6% of concerns fall within their fine region's majority coarse parent. Neither level of detail identifies a unique grouping.

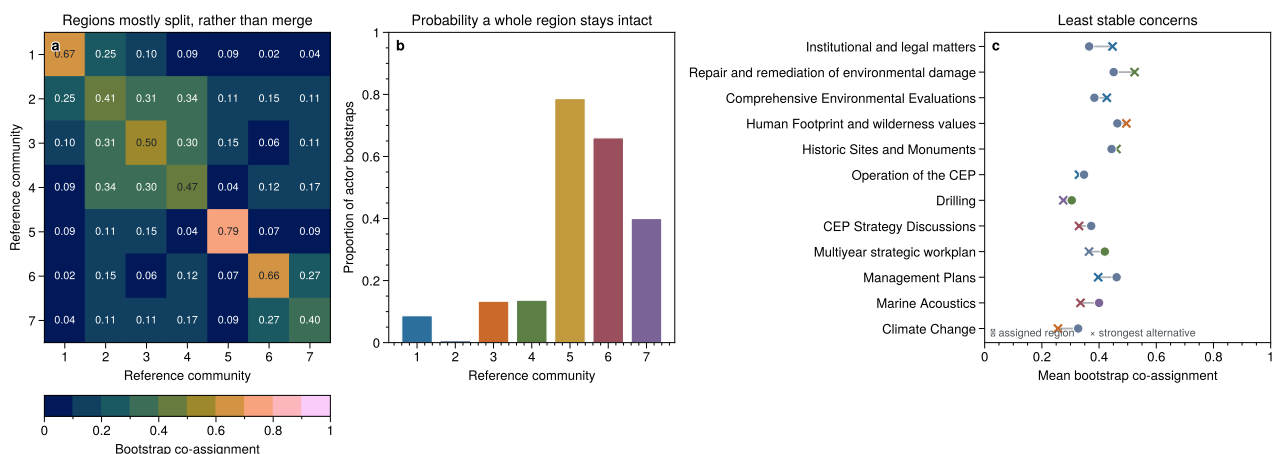


Figure S8: Actor bootstraps split the seven displayed regions more often than they merge them. Each of 300 resamples rebuilds RPA, shared-specialization scores, and the Constant Potts Model grouping at $\gamma = 0.273$. (a) Mean frequency with which pairs enter the same region, shown within displayed regions (diagonal) and between them (off-diagonal). (b) Probability that an entire region remains intact. (c) Twelve concerns with the smallest gap between placement with their displayed region (circle) and strongest alternative (cross).

and management". Table S3 shows the complete alignment. scripts/official_regular_atcm_outputs.py freezes it, and scripts/analyze_output_category_families.py rebuilds the broad paper map and instrument analysis.

We audited the category human mapping with one author-coded blinded to the author-prediction results. The coder received only the 45 official instrument labels, the 15 categories in order, and an instruction to select each functional principal institutional function. The coder matched

38 of 45 (84.4%) author-coded segments recorded high confidence for the seven CEP Discussions, Drilling, Treaty Biological Inspections, the CEP. We of the author-coded and treat these

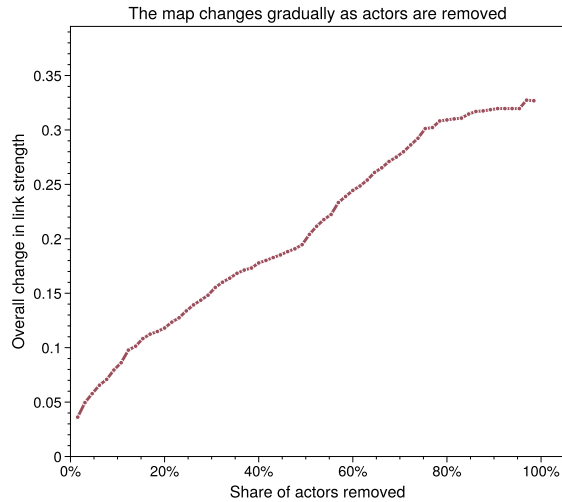


Figure S9: No single actor determines the concern map. Each step removes the actor whose exclusion maximizes root mean squared change from the original 66-actor shared-specialization matrix, then rebuilds the map. Change accumulates gradually. We omit the undefined comparison after the final removal.

bels as boundary cases in the mapping-sensitivity analysis. tools/build_crosswalk_blind_validation.py creates the blinded packet, and output/scientific_checks/crosswalk_blind_coder_a_comparison records the item-level audit.

We test dependence on this mapping using 27 boundary concerns and 50 plausible alternative assignments based on the concern labels. Each one-at-a-time test rebuilds paper-to-category relations, historical maps, papers about the target category, and papers weighted by nearby categories. We also evaluate 48 sparse joint mappings, in which each boundary concern has a 20% reassignment probability, and 48 broad joint mappings drawn uniformly from the retained options. These ensembles generate stress scenarios rather than probabilities for the correct mapping. A post hoc failure search combines the individually most adverse assignments. Because it selects assignments using evaluation scores, it locates vulnerability rather than expected performance under coding error. Official instrument categories, evaluation meetings, model settings, and scoring remain fixed. scripts/check_crosswalk_uncertainty.py reproduces these checks.

A.12 Retrospective predictions of adopted text categories

We test whether papers help locate the categories of adopted texts. The target is not the total number of instruments. For each form of output and each evaluation meeting in sequence, it is the distribution of that meeting's instrument weight across the 15 official categories. If instrument r at meeting t assigns observed share q_{irt} to category i , and a model assigns predicted share $\hat{p}_{irt}$, we score the prediction as

$$\mathcal{L}_{rt} = -\sum_i q_{irt} \log \hat{p}_{irt}. \quad (7)$$

We call $\mathcal{L}$ the allocation log score. Lower values mean that the model placed more probability on the categories where the meeting adopted output. We refer to it as the prediction score in the reader-facing results.

We select the specification using Measures, Decisions, and Resolutions together from ATCMs 25–28. Candidate settings include

windows of 3, 5, 8, or 10 meetings of earlier outputs and windows of 1, 2, 3, or 5 meetings of papers about the target category. We also compare regularization values of 0.1, 0.2, 0.5, and 1.0. This procedure selects five meetings of earlier outputs, one meeting of earlier papers about the target category, and regularization 0.1. We freeze those settings and run a sequence of predictions trained on earlier meetings for ATCMs 29–47, giving 19 evaluation meetings for each form of output. At each evaluation meeting, we fit the model on ATCM 20 through the meeting immediately before it. A regularized Poisson model includes category indicators and standardised numerical predictors. We normalise predictions across categories before scoring. Earlier exploratory analyses had already examined ATCMs 29–47, so this period provides a retrospective test trained only on earlier meetings instead of external validation.

The output-history model contains the preceding five meetings of the same form of output in the target category. The target-category paper model adds fractionally counted papers and the number of distinct submitting actors in that category during the immediately preceding meeting, plus the same two quantities for the target meeting. Participants tabled the paper files before meeting outputs, but the category memberships used to aggregate them come from 2026 query responses. Because those responses lack assignment timestamps, we treat the exercise as a retrospective same-meeting prediction. The full model adds papers from the target meeting in every non-target category with a positive relation, up to 14. For each target category, we normalise shared-specialization scores from the earlier-paper map to one and apply them to paper counts from the target meeting. This addition uses every positively related category and therefore requires no chosen number of neighbours. We rebuild the broad map from papers submitted before the target meeting.

We compare models within each evaluation meeting and resample the 19 paired score differences 20,000 times to form 95% intervals. Contrasts among forms of output resample the paired difference between Resolution gains and the corresponding Measure or Decision gains. Robustness checks vary the earlier-output window, target-paper window, regularization, and number of neighbours. They also omit each target category from scoring, replace papers about nearby categories with all other papers, and project current papers through the original 45-concern map. Further checks split the evaluation period and use moving-block meeting bootstraps. A separate check reselects every setting using only meetings before each evaluation meeting.

As a post hoc concentration check, we calculate the adopted Shannon entropy of the within Resolution distribution meeting and split the 19 meetings at the median. We report the mean score change in each half, their difference, and entropy. In-tervals resample each half. This check asks whether the gain appears only in meetings with unusually broad Resolution output; it does not make Measures, Decisions, and Resolutions compositionally equivalent. scripts/analyze_resolution_concentration_sensi reproduces the calculation.

The shuffled-map comparison uses 200 label permutations. At each meeting, it applies the same permutation to the rows and columns of the matrix built from earlier papers, recomputes the paper counts weighted by nearby categories, and refits every

Table S3: An author-coded mapping assigns paper concerns to 15 official instrument categories. The left column lists categories from the Antarctic Treaty System instrument register. The right lists the paper concerns assigned to each. The mapping is unofficial.

Official instrument category	Included paper concerns
Area protection and management	Area Protection and Management Plans General; Management Plans; Environmental Domains Analysis; Marine Protected Areas
Institutional & legal matters	Operation of the ATS: General, Reports, and Secretariat; Cooperation with Other Organisations; Liability; Institutional and legal matters; Multiyear strategic workplan
Tourism and Non-Governmental Activities	Tourism and NG Activities; Site Guidelines for Visitors
Operational matters	Inspections; Search and Rescue; Operational issues; Safety and Operations in Antarctica; Emergency report and contingency planning
Environmental protection	Climate Change; Environmental Protection General; Environmental Monitoring and Reporting; CEP Strategy Discussions; Human Footprint and wilderness values; Operation of the CEP; State of the Antarctic Environment Report; Repair and remediation of environmental damage
Historic Sites and Monuments	Historic Sites and Monuments
Information exchange	Exchange of Information
Scientific cooperation	Science issues; Marine Acoustics; International Polar Year; Sub-glacial Lakes; Drilling; Biological Prospecting
Fauna and flora	Fauna and Flora General; Specially Protected Species; Nonnative Species and Quarantine
Marine pollution	Prevention of marine pollution
Environmental impact assessment	Environmental Impact Assessment: Other EIA Matters; Comprehensive Environmental Evaluations
General matters	Educational issues; Opening statements
Marine living resources	Marine living resources
Waste disposal and management	Waste management and disposal
Mineral resources	Mineral resources

prediction. The test therefore breaks which categories occupy neighbouring positions while preserving a coherent symmetric map. Because lower score changes are better, its lower-tail probability is $(1+b)/(1+200)$, where b counts shuffled changes less than or equal to the observed change. The Area protection and management check omits that target row from scoring but leaves its papers available as attention to nearby categories. `scripts/analyze_resolution_attention_forecast.py` reproduces the primary predictions and checks. The script in `scripts/evaluate_nested_attention_forecast.py` reproduces the within-fold selection sensitivity.

Three post hoc checks probe the use of information unavailable at the time and the persistence of attention. The specification using only earlier information excludes paper counts, actor reach, and attention to nearby categories from the target meeting. It uses only the corresponding quantities from the preceding meeting. Moving-block intervals resample blocks of one to four meetings, and an exact sign-flip test covers all 2^{19} paired sign assignments. A check using later information replaces predictors for the target meeting with the next meeting's attention for the 18 meetings with available data. We also rebuild the analysis for ATCMs 29–45 from the January 2024 single-category snapshot and compare it with the current multi-label reconstruction over the same meetings. `scripts/adversarial_forecast_checks.py` and `scripts/audit_snapshot_nowcast_sensitivity.py` reproduce these tests.

A.13 Auditing formal citations among Measures

The prediction asks whether attention helps locate adopted output. A separate audit asks what formal record later Measures cite. We froze the adopted body text from the official detail page of every one of the 277 Measures on 16 August 2026. The pinned JSON Lines (JSONL) corpus records each official URL, retrieval date, extraction status, and body hash; an adjacent manifest records the corpus SHA-256 and verifies the source inventory and all 277 records. The audit consumes this corpus without live requests. We extract exact references of the form *Measure*, *Decision*, or *Resolution* followed by a number and year. Instrument numbers restart each year, so type, number, and year jointly identify every source. A reference counts only when it resolves to an earlier entry in the pinned official inventory. Repeated mentions of the same source within one Measure count as one link.

The audit reports the number of later Measures citing each source type, the number of distinct earlier instruments cited,

and the number of source–Measure links. A later Measure can cite several source types, so the percentages in fig. 3b overlap. These exact references provide a conservative record of formal citation. They do not establish that the earlier instrument caused the later one, and a policy sequence can omit an intermediate source from its final adopted body text. We also screen Measure titles for “management plan” and “revised management plan” to describe the recurring content of the Measure record. `scripts/plot_formal_lineage_overview.py` reproduces the lineage overview.

To examine whether heavily discussed Resolutions later receive a citation from a Measure, we retain Resolutions with ten subsequent regular meetings in the archive. A Resolution enters the linked set only when a Measure adopted within those ten meetings contains an exact year-qualified citation to it. For each linked Resolution, we compare same-category paper attention with uncited Resolutions sharing at least one official category and report its percentile within that set. We count time in meetings rather than calendar years. The linked sample is small, so this is a descriptive boundary test rather than a fitted transition model. `scripts/audit_resolution_measure_ladders.py` and `scripts/analyze_resolution_hardenening_attention.py` reproduce the audit.

A.14 Paper–output associations

The category–meeting analysis orders the record by ATCM number, so one step is the immediately preceding meeting even when more than one calendar year elapsed. For broad category i at ATCM t , we measure three earlier histories: fractionally counted papers in i ; papers in its five nearest broad categories, weighted using the map available before each source meeting; and adopted outputs in i . For the primary one-meeting analysis, each history comes from ATCM $t-1$, and its nearby-paper weights use the map built through $t-2$. We add one and take the logarithm.

A Poisson pseudo-maximum-likelihood model uses category and meeting fixed effects. The reported rate ratios give the change associated with doubling one plus an earlier count. We cluster covariance by category and meeting. Because there are only 15 category clusters, the intervals require caution. They also condition on the author-coded category mapping and estimated map; the models do not propagate uncertainty in those objects.

We repeat the analysis after combining the preceding one, two, three, five, eight, or ten meetings. The available meeting set therefore shrinks as the window grows. To separate window length from sample period, we repeat every horizon

on the 19 meetings shared by the ten-meeting model (ATCMs 29–47). Composition checks exclude General matters, Area protection and management, or both. The latter matters because recurring area and management-plan work makes up almost half of the adopted-output category weight. All comparisons describe temporal association between documentary streams and adopted outputs, not a causal effect of papers on output. `scripts/analyze_output_category_families.py` reproduces the models.

B Supplementary Results

B.1 Prediction performance differs among forms of output

The repeated comparison trained on earlier meetings separates papers about the target category from papers about nearby categories (table S4). Together they improve the Resolution prediction relative to the categories addressed by earlier Resolutions. The intervals for the two components considered separately cross zero. The combined model and its target-category component show no clear improvement for the Measure or Decision prediction. The complete Resolution gain is larger than the Measure gain by 0.063 ($[-0.112, -0.012]$) and larger than the Decision gain by 0.092 ($[-0.159, -0.022]$).

Later Measures cite nine source Resolutions. Across these sources, the first Measure to cite each Resolution followed it by a median of four Antarctic Treaty Consultative Meetings (range, 1 to 16 meetings; $n=9$). Later Measures cited these Resolutions over decades (fig. S10b). Eight of the nine source Resolutions and 71 of the 73 citing Measures carry the official Area protection and management category. Each source Resolution and the later Measures that cite it form a formal citation lineage. These nine lineages document repeated formal citations concentrated in one policy area, with three citation links outside it; they do not establish a general mechanism in which Resolutions lead to Measures.

The three forms of output pose different prediction problems. The effective number of occupied categories, calculated as the exponential of Shannon entropy, is 10.55 for Resolutions, 3.14 for Decisions, and 1.38 for Measures. Area protection and management accounts for 93.1% of fractional Measure weight, while Institutional and legal matters accounts for 73.0% of Decision weight. The largest Resolution category, Tourism and Non-Governmental Activities, accounts for only 22.1%. Earlier output is therefore already a strong guide to the narrow Measure and Decision distributions.

Within the 19 Resolution evaluation meetings, the combined score change is -0.073 in the ten meetings at or below median category entropy ($[-0.141, -0.006]$) and -0.055 in the nine meetings above it ($[-0.125, 0.021]$). Their difference is 0.018 ($[-0.080, 0.119]$). The correlation between score change and entropy is -0.043 ($[-0.417, 0.318]$). The point estimates remain similar across narrower and broader Resolution meetings, but the intervals remain wide.

For Measures, the output-history model based on the preceding five meetings of Measures gives a mean prediction score of 0.388. Models using prior Decisions and Resolutions alone score 1.134, all earlier output scores 0.486, and papers about the target and nearby categories without Measure history score 0.759. Adding Decisions and Resolutions to the Measure output-history model worsens the score by 0.011 ($[0.000, 0.023]$). Adding papers to that model changes it by -0.001 ($[-0.023, 0.020]$). The recurrence is consistent with the content of the record: 266 of 277 Measures carry the Area protection and management category, 230 titles concern a management plan, 211 concern a revised management plan, and 215 explicitly identify an ASPA (fig. S11).

The Resolution result is not reproduced by generic activity outside the target category. Adding all other papers to the

target-category paper model changes the score by 0.001 ($[-0.001, 0.003]$), compared with -0.023 for papers weighted by closeness to the target. Omitting each broad category from the score leaves the mean gain from nearby categories negative in all 15 cases, ranging from -0.033 to -0.016 . Every one of the 16 combinations of earlier-output and paper windows gives a negative mean gain, as do all five tested neighbour counts and all four regularization values. All six projections through the original 45-concern map improve the mean score. Moving-block bootstraps of one to four meetings preserve the negative mean, although their intervals cross zero. The gain is -0.008 in ATCMs 29–38 ($[-0.038, 0.025]$) and -0.040 in ATCMs 39–47 ($[-0.083, -0.005]$). When every setting is instead reselected within each training period using earlier meetings, the combined Resolution change remains similar at -0.067 , but its interval crosses zero ($[-0.131, 0.004]$). We use the fixed early-period specification as the primary summary because its settings are transparent and shared across forms of output; the reselection result shows that model selection adds uncertainty without reversing the mean pattern.

B.2 Timing and category-mapping provide bounds for prediction

Papers from the preceding meeting improve only the Resolution prediction. Papers about the target category change the Resolution score by -0.042 ($[-0.076, -0.006]$), and papers about the target and nearby categories change it by -0.082 ($[-0.128, -0.033]$). The combined model improves 15 of 19 meetings, and moving-block intervals of one to four meetings remain below zero. The exact two-sided sign-flip probability is 0.005. For Measures and Decisions, the corresponding preceding-meeting changes are 0.007 ($[-0.022, 0.043]$) and 0.017 ($[-0.025, 0.058]$). The result therefore retains the separation among forms of output.

Future attention provides a weaker persistence check. Attention from the next meeting changes the Resolution score by -0.055 across 18 meetings and improves 12. Moving-block intervals for blocks of one and two meetings cross zero, and the exact two-sided sign-flip probability is 0.115. This result does not separate a stable agenda from directional prediction, but the clearer preceding-meeting association shows that focal-meeting attention is not necessary for the main pattern.

The January 2024 snapshot preserves one category for 6,230 papers that also appear in the 2026 reconstruction. Every retained category matches its later record, including all 3,644 matched papers from ATCMs 29–45. Over these 17 evaluation meetings, the older labels preserve the target-paper and combined Resolution gains. The improvement from nearby categories changes from -0.018 ($[-0.043, 0.006]$) under the multi-label reconstruction to 0.003 ($[-0.002, 0.008]$) under the single-label snapshot. The snapshot cannot date its labels and excludes ATCMs 46–47. It supports stability of the overall attention association but weakens attribution to attention from nearby categories alone.

Changes to the category mapping identify a separate boundary. All 50 one-at-a-time reassignments retain a negative estimate for papers about the target and nearby categories, ranging from -0.075 to -0.049 , and 48 intervals remain below zero. All 48 sparse joint mappings retain negative estimates from -0.085 to -0.029 , with 34 intervals below zero. All 48 broad joint mappings also retain negative estimates from -0.060 to -0.008 , with 15 intervals below zero. The improvement from nearby categories remains negative in every isolated and sampled mapping, although its interval often crosses zero. A post hoc stack of 19 individually adverse assignments moves the combined estimate to 0.009 ($[-0.033, 0.053]$) and the improvement from nearby categories to 0.001 ($[-0.004, 0.006]$). Local and sampled mapping uncertainty does not explain the main point estimate, but a coordinated alternative mapping can remove it. The independent blinded AI coder agrees with 84.4% of the author-coded mapping and locates seven

Table S4: Papers about target and nearby categories improve retrospective Resolution category predictions, while their gains over earlier output are unclear for Measures and Decisions. Entries are mean prediction-score changes across 19 evaluation meetings assessed in sequence (ATCMs 29–47), with meeting-bootstrap 95% intervals. Lower scores are better. The first two columns compare papers about the target category and papers about the target plus nearby categories with the preceding five meetings of output categories. The final column compares the target-plus-nearby model with the target-only model. The target is category shares conditional on observed output. The analysis draws category metadata from a retrospective archive, and earlier exploratory analyses included the evaluation meetings. Reselecting the settings separately for each evaluation places the combined Resolution interval across zero.

Form of output	Target papers vs earlier output	Target + nearby papers vs earlier output	Nearby contribution
Measure	0.007 [−0.012,0.025]	−0.001 [−0.023,0.020]	−0.008 [−0.015,−0.001]
Decision	0.024 [−0.023,0.069]	0.028 [−0.022,0.075]	0.003 [−0.002,0.009]
Resolution	−0.041 [−0.082,0.001]	−0.065 [−0.114,−0.014]	−0.023 [−0.050,0.002]

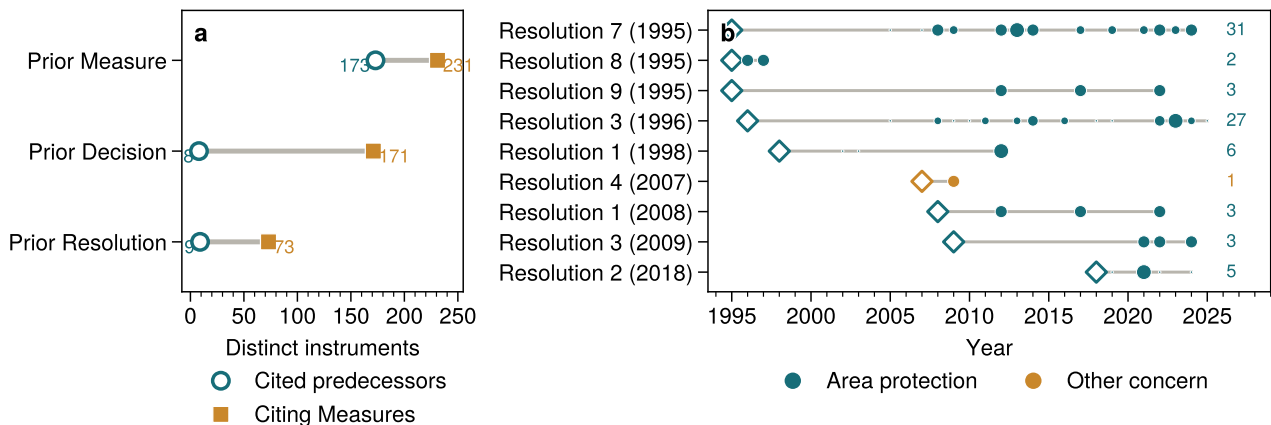


Figure S10: Exact citations link later Antarctic Treaty System Measures to earlier Measures, Decisions, and Resolutions. a, 231 distinct later Measures cite 173 distinct earlier Measures, 171 distinct later Measures cite at least one of eight Decisions, and 73 distinct later Measures cite at least one of nine Resolutions. Rows overlap because a Measure can cite several source types. b, each row traces one cited Resolution. The diamond marks its adoption year, and circles mark later years when Measures cite it; circle area scales with the number of citing Measures in that year. The number at right gives the number of distinct Measures that cite that Resolution. Eight of the nine source Resolutions carry the official Area protection and management category; teal marks those eight lineages, and orange marks the other lineage. Across all nine lineages, 71 of the 73 distinct citing Measures also carry that category. Exact citations establish formal citation links. They do not show that a cited instrument caused a later Measure or that any Measure entered into force.

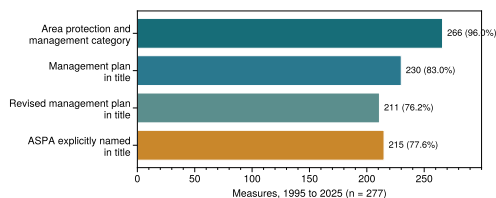


Figure S11: Measures concentrate in recurring area-protection and management-plan work. Bars show overlapping indicators among 277 Measures adopted at regular ATCMs from 1995 to 2025. Area protection and management is an official instrument category; the remaining indicators come from Measure titles. The ASPA bar combines 160 titles that spell out “Antarctic Specially Protected Area” and 55 that use the acronym.

disagreements among the boundary concerns examined here.

B.3 Measures cite earlier Measures most widely

The adopted body text of 245 of 277 Measures cites at least one earlier Measure, Decision, or Resolution (table S5). Prior Measures provide the broadest source set: 173 distinct earlier Measures appear in 474 links to 231 later Measures. The 261 Decision links come from only eight distinct Decisions, showing repeated use of a small administrative foundation. Nine distinct Resolutions appear in 81 links to 73 Measures.

Seventy-eight of the 81 Resolution-to-Measure citation links have a target Measure in the official Area protection and management category; two further links concern heritage work. Repeated Measures cite Resolutions that set expiry rules, management-plan formats, revision responsibilities, or heritage-assessment procedures. Resolution 4 (2007) on ship-based tourism and Measure 15 (2009) on passenger-vessel landings form the only cited pair outside those domains.

Exact citations miss intermediate steps that the final Measure does not repeat. Decision 3 (1998) moved further negotiation of a liability annex to ATCM Working Group I, and Resolution 5 (1999) requested practical and scientific input. Decision 3 (2001) advanced a draft, Measure 1 (2005) adopted Annex VI, and Decision 1 (2005) established annual review of progress towards its becoming effective. The 2005 Measure does not cite the 1999 Resolution. For this reason, table S5 is a lower bound on documented continuity and not a complete reconstruction of instrument development.

Requiring ten subsequent meetings leaves 115 eligible Resolutions; Measures adopted within that horizon cite six. Their same-category paper attention is not unusually high relative to uncited Resolutions sharing an official category. The linked Resolutions average the 24th percentile for the current and two preceding meetings, then the 33rd, 36th, 39th, 42nd, and 43rd percentiles after one, three, five, eight, and ten meetings. The sample is too small for a general transition model, but it does not support a simple rule in which Measures later cite the most heavily discussed Resolutions. A later Measure citation also does not establish that the Measure entered into force.

Table S5: Measures cite earlier Measures more widely than Decisions or Resolutions. The inventory contains 277 Measures from regular ATCMs, 1995–2025. Links require an exact, year-qualified reference in a later Measure's adopted body text and a match to the official inventory. A Measure can cite several source types, so rows overlap. Lags are calendar years.

Earlier instrument form	Distinct later Measures	Distinct earlier outputs	Citation links	Median lag
Measure	231	173	474	8
Decision	171	8	261	11
Resolution	73	9	81	18

B.4 The paper–output association changes with sample period

At the immediately preceding ATCM, doubling one plus papers in the same broad category corresponds to an output-rate ratio of 1.189 ([1.036,1.364]). The corresponding estimate for papers in the five nearest categories is 0.989 ([0.926,1.055]), and that for earlier output in the same category is 1.396 ([1.162,1.677]). The ratio between the same-category and nearby-paper rate ratios is 1.202 ([1.023,1.413], $p=0.025$; fig. S12a). The nearby estimate is conditional on same-category papers and should not be read as evidence that neighbouring attention suppresses output.

Changing the window also changes which meetings can enter the model. With every eligible meeting included, the same-category versus nearby contrast is clear at one meeting (1.202, [1.023,1.413]) and three meetings (1.304, [1.032,1.648]), but not at five meetings (1.391, [0.936,2.067]). Over the same 19 later meetings, the corresponding contrasts are 0.922 ([0.699,1.216]), 0.961 ([0.671,1.375]), and 0.979 ([0.577,1.662]). None of the estimates over the same 19 meetings is clear (fig. S12b). The full-record result therefore varies with sample period rather than identifying a fixed lag or cumulative conversion law.

Composition changes the precision of the result. After removing Area protection and management from the paper histories, broad map, and output panel, the one-meeting contrast is 1.309 ([0.997,1.719]); after also removing General matters, it is 1.230 ([0.853,1.774]). The point estimates do not shrink, so these checks do not show that area management creates the association. They do show that the full-record contrast is no longer clear once we remove the dominant category. Almost half of output category weight belongs to Area protection and management, much of it recurring site and management-plan work (fig. S12c).

B.5 Consultative Parties preserve the main patterns

We repeated the concern-map, portfolio-entry, and adopted-text prediction analyses after restricting documentary attention to Consultative Parties. We used recognition dates from the official ATS Parties table and treated a state as eligible in its recognition year and every subsequent meeting year. Papers submitted only by Non-Consultative Parties, observers, invited expert organizations, or the Secretariat did not enter this analysis. For jointly submitted papers, only eligible Consultative Parties received actor-level credit, while each retained paper contributed one total unit of attention across its archive categories. All other definitions matched the main analyses.

The restricted corpus contains 5,192 papers attributed to 29 Consultative Parties. All 45 concerns remain connected, and pairwise proximities correlate with the all-actor reconstruction at Pearson $r=0.742$ and Spearman $\rho=0.730$ (fig. S13a). Portfolio entry remains local: each 0.1 increase in distance corresponds to an odds ratio of 0.900 (95% interval [0.860,0.942]) across 741 actor-period comparisons (fig. S13b). For Resolutions, the target-and-nearby attention model lowers the mean prediction score by 0.050 relative to the output-history model (95% meeting-bootstrap interval $[-0.100,-0.003]$) and scores better in 13 of 19 meetings. The corresponding intervals for Decisions and Measures include zero (fig. S13c). The connected map, local portfolio development,

and the contrast among forms of output therefore persist when the documentary record is restricted to formal decision-makers.

B.6 Cohort positions differ during the first 15 years

In a retrospective comparison using the full-record map, actors joining in 1991–2010 ($n=17$) built opening portfolios around concerns whose specialized actors held fewer concerns across the full record than expected ($0.929\times$ the opening-window benchmark; 95% actor-bootstrap interval [0.889,0.966]). Those concerns also have fewer neighbours in that map on average ($0.883\times$ the benchmark, [0.798,0.974]). Earlier cohorts sit near parity on both measures. The full estimates and construction are in section A.6 and table S2. This selected comparison excludes post-2010 entrants by design, conditions eligibility on a 15-year record, and does not reconstruct the information set available at entry.

B.7 Alternative matched comparisons

The paired test compares each realized specialization shift with a size-matched redraw from concerns the actor did not hold. We vary the full-record map or maps built from all earlier records, uniform or popularity-weighted redraws, and whether we draw candidates from all eventual labels, those observed by the current-window end, or those observed before the transition. Popularity weights equal the number of actors specialized in the concern in the previous window plus one. Each event uses 200 redraws.

Every construction places re-
alized shifts nearer than its
matched alternatives more of-
ten than not. The pri-
mary comparison uses the ear-
ly map built from all ear-
lier records, and only prior popular-
ity, observed before the tran-
sition. It contains 968 actor-
periods and gives 62.5%
nearer (95% actor-bootstrap in-
terval [58.9, 66.5%]). The
observed median displacement
is 0.576, against a
weighted-alternative median
of 0.594 ($p=1.3\times 10^{-15}$ in the unclustered
Wilcoxon comparison). scripts/compare_major_results_category_treatment
reproduces the comparison and actor bootstrap.

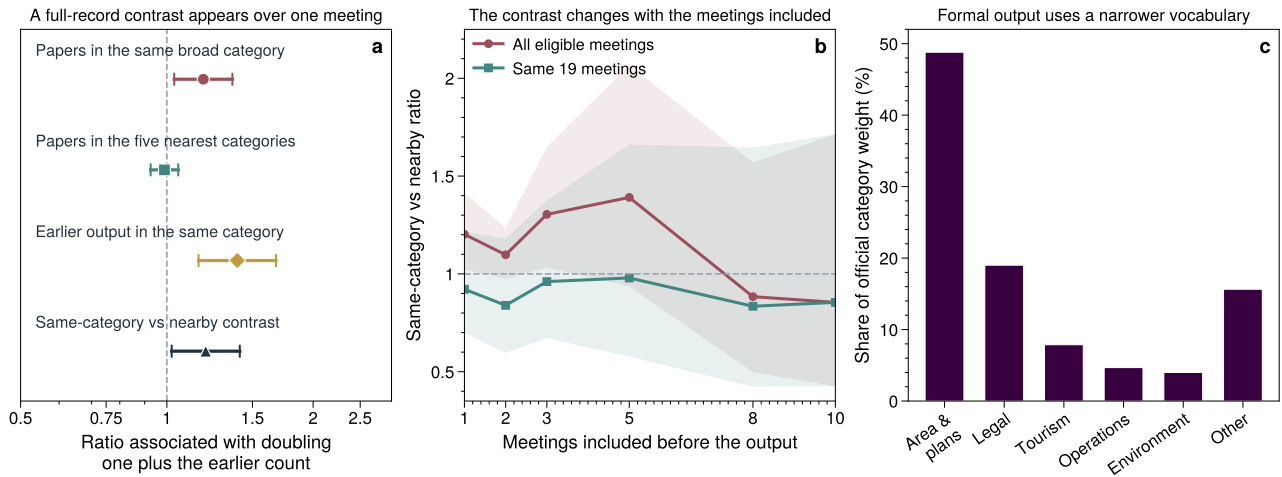


Figure S12: The paper-output association across forms of output changes with sample period. (a) Output-rate ratios associated with doubling one plus an earlier count from the immediately preceding Antarctic Treaty Consultative Meeting (ATCM). The fourth estimate is the ratio of the same-category and nearby-paper rate ratios. Bars are category- and meeting-clustered 95% intervals. (b) The same ratio across windows of 1, 2, 3, 5, 8, and 10 meetings, using all eligible meetings or the common 19-meeting sample. (c) Fractional category weight for 584 outputs across 15 categories. This boundary test combines all three forms of output and differs from predictions fitted separately for each form.

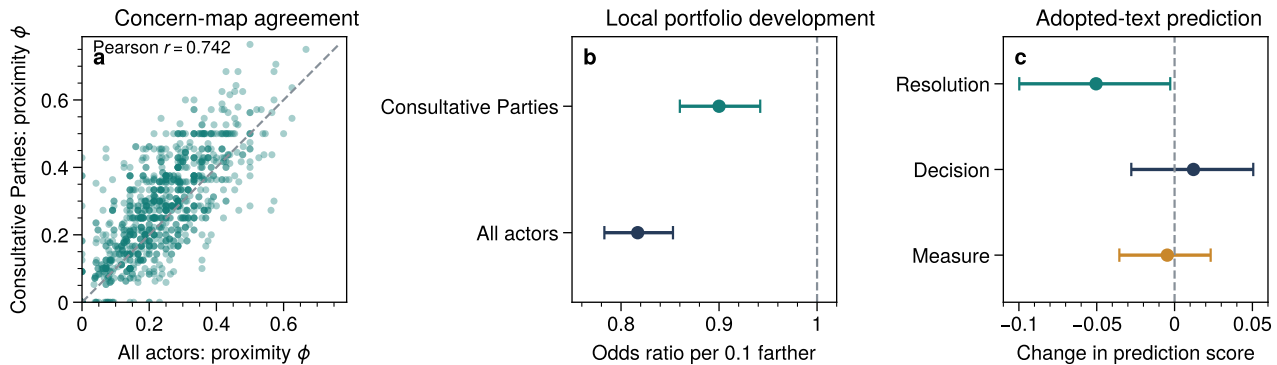


Figure S13: Consultative Parties preserve the connected map, local portfolio development, and Resolution prediction. (a) Pairwise concern proximity in the full actor set and among Consultative Parties only. The dashed line marks equality. (b) Odds ratios for entering a specialization 0.1 farther from the earlier portfolio. Points show estimates and bars show 95% intervals. (c) Change in prediction score when the target-and-nearby attention model is compared with the output-history model in the Consultative-Party-only analysis. Negative values indicate improved prediction; bars show meeting-bootstrap 95% intervals.

Table S6: Local specialization persists across alternative definitions and timing choices. Rows vary the comparison, outcome, category assignment, or time window. OR is the odds ratio for entering a specialization after a 0.1 increase in distance from the earlier portfolio. Values below one indicate locality. The first row uses an actor-bootstrap interval, and OR rows use model-based intervals. In the combined robustness row, estimates report the first-paper outcome followed by one-category assignment. The final row reports duration under alternative lapse rules and has no OR.

Comparison	Result	Question answered
Popularity-weighted alternatives using only earlier information ($n = 968$)	62.5% nearer, actor-bootstrap interval [58.9, 66.5]%	Is the realized specialization shift nearer than alternatives weighted by their earlier popularity?
Within-actor model using only earlier information (30,957 actor-period-concern options)	OR 0.817 [0.783, 0.853] per 0.1 farther	Among options open to the same actor at the same time, are nearby concerns more likely to cross the threshold?
First-paper outcome and one-category robustness	OR 0.874 [0.834, 0.916] and 0.831 [0.794, 0.870]	Does the direction remain after changing the outcome or forcing one inferred category per paper?
Disjoint five-meeting windows	OR 0.904 [0.855, 0.957]	Does the adjacent-window result extend to non-overlapping blocks?
Years active after a concern first enters	5–42% remain after ten years and fewer than 20% after twenty years, depending on whether the rule allows a lapse of one to five years	How much does the length of apparent specialization depend on the rule for a temporary lapse?

B.8 A boundary test: the 1991 Madrid Protocol

To ask whether pre-shock position also predicts responses outside routine portfolio evolution, we examine the reorientation surrounding the 1991 Madrid Protocol. Parties adopted the Convention on the Regulation of Antarctic Mineral Resource Activities (CRAMRA) in Wellington in 1988 and abandoned it after Australia and France declined to sign it in 1989; the Protocol negotiations of 1989–91 replaced it [23]. The documentary record in 1986–90 therefore contains attention to the failed instrument as well as the concerns later reorganized by the Protocol.

The design uses pre-1991 inputs where possible. We first restrict eligibility to concerns representing at least 1% of the pre-1991 paper record, which excludes concerns created by the Protocol. Among the eligible concerns, the post-Protocol growth set contains the eight whose archive share grew most between 1986–90 and 1992–96. Using documents from 1961–90, we measure how close an actor's pre-1991 portfolio ($RPA \geq 1$) lies to the post-Protocol growth set by its highest shared-specialization score. The outcome measures the share of an actor's papers in each 5-year window that falls in the post-Protocol growth set. We model it with window dummies interacted with this closeness measure, actor and window fixed effects, window-by-portfolio-size controls, and actor-clustered standard errors. The model covers 30 actors and 263 actor-window observations with at least three papers.

The pre-period coefficients do not reject parallel trends (joint $F(5, \cdot) = 1.61$, $p = 0.159$), although this low-powered diagnostic does not establish that the shock is exogenous. Every post-shock closeness-by-window coefficient is negative and small relative to its uncertainty (range -0.83 to -0.19 percentage points). A second-pass model asks whether relative closeness to the post-Protocol growth set predicts entry into a predefined six-concern Protocol set: Liability, Inspections, Environmental Monitoring and Reporting, Environmental Protection (General), Waste Management and Disposal, and Comprehensive Environmental Evaluations. Its coefficient is also negative and uncertain (-3.02 , SE 3.12 , $p = 0.334$). The closeness measure correlates strongly with pre-1991 portfolio size ($r = 0.87$), which the window-by-portfolio-size controls absorb.

The negative, imprecise estimates leave the association under an unusually large institutional reorientation unresolved. Pre-shock position did not clearly distinguish who responded, and the design cannot determine whether the collective response overrode a normally local process or arose through a different process altogether. We report this exploratory boundary test because its outcome limits the mechanism reading of the main text. `scripts/madrid_shock_exploration.py` reproduces the analysis.